Tectonophysics 779 (2020) 228386

Contents lists available at ScienceDirect

Tectonophysics

journal homepage: www.elsevier.com/locate/tecto

Paleogeographic and tectonic evolution of the Pampia Terrane in the Cambrian: New paleomagnetic constraints

Pablo R. Franceschinis[a,b,*], Augusto E. Rapalini[a,b], Mónica P. Escayola[b,c], Constanza Rodríguez Piceda[d,e]

[a] Universidad de Buenos Aires, Laboratorio de Paleomagnetismo Daniel A. Valencio, Instituto de Geociencias Básicas, Aplicadas y Ambientales de Buenos Aires (IGEBA), Departamento de Ciencias Geológicas, Facultad de Ciencias Exactas y Naturales, Argentina
[b] CONICET, Argentina,
[c] ICPA-Universidad Nacional de Tierra del Fuego, Antártida e Islas del Atlántico Sur, Argentina
[d] Institut für Geowissenschaften, Universität Potsdam, Potsdam, Germany
[e] GFZ – German Research Center for Geosciences, Potsdam, Germany

ARTICLE INFO

Keywords:
Campanario Formation
Puncoviscana Formation
Pampia Terrane
Gondwana

ABSTRACT

Paleomagnetic, magnetic fabric and rock magnetic studies were carried out in the late Middle to early Late Cambrian Campanario Formation exposed in NW Argentina. The study also presents preliminary results from the Early Cambrian metasediments of the Puncoviscana Formation and from Mojotoro intrusive. A new paleomagnetic pole, C2, was computed for the Campanario Formation (23.6°N, 346.5°E, A95, 7.0°, N: 11 sites) from the localities of Tilcara (T), Terma de Reyes (R) and El Perchel (EP), including previous data from Santa Victoria Oeste (M), in northernmost Argentina. A positive fold test was obtained for Tilcara locality while a positive regional tilt test was obtained for mean site directions for the four localities. This pole can be considered as the representative pole position for the Pampia terrane for the late Middle to Late Cambrian. Previous paleomagnetic data for the same formation from the Iruya-Matancillas locality show an *in situ* clockwise rotation of about 30° around a vertical axis when compared with the mean Pampia pole. The same is observed when comparing the early Ordovician pole for the Santa Rosita Formation at those localities with the recently obtained coeval pole of the La Pedrera Formation, indicating that local tectonic rotations, affected that area. These results suggest that previous paleomagnetic interpretations for these Late Cambrian and Early Ordovician rocks in the region as recording displacement of the Pampia terrane in Cambrian times are probably incorrect. Additionally, two virtual geomagnetic poles were computed for the Puncoviscana Formation (TP1: 34.5°N, 52.3°E, A95: 5.0°, n: 22 specimens and TP2: 38.3°N, 33.6°E, A95: 5.7°, n: 14 specimens) and one for the Mojotoro Intrusive (MO: 9.1°N, 345.6°E, A95: 16.4°, n: 11 specimens). These and previous Early Cambrian paleomagnetic data from the Pampia terrane differ from Gondwana coeval reference directions in inclination and/or declination, lacking a simple pattern.

1. Introduction

Our understanding of the Early Paleozoic paleogeographic evolution of the Pampia Terrane (see Ramos et al., 2010) is still ambiguous and has been interpreted with diverse tectonic and kinematic models (Schwartz and Gromet, 2004; Rapela et al., 2007; Ramos et al., 2010; Escayola et al., 2007, 2011; Spagnuolo et al., 2012). Paleomagnetic results from the Middle to Upper Cambrian Campanario Formation (Spagnuolo et al., 2008, 2012; Franceschinis et al., 2016) and from the early Ordovician Santa Victoria Group (Spagnuolo et al., 2012; Rodríguez Piceda et al., 2018) have provided some constraints for this problem but could not solved ambiguities in the tectonic interpretation of such results. Knowledge of the mechanism and time of accretion of Pampia to Gondwana would place important paleogeographic restrictions on the processes that led to final assembly of Gondwana and their chronology. The existence of the hypothetical Clymene ocean in late Ediacaran times (Trindade et al., 2006; Cordani et al., 2013; Tohver and Trindade, 2014), between the Amazonian craton and the Pampia terrane on one side and the Rio de la Plata and Congo-São Francisco cratons on the other, is still a matter of controversy. According to some

* Corresponding author at: Universidad de Buenos Aires, Laboratorio de Paleomagnetismo Daniel A. Valencio, Instituto de Geociencias Básicas, Aplicadas y Ambientales de Buenos Aires (IGEBA), Departamento de Ciencias Geológicas, Facultad de Ciencias Exactas y Naturales, Argentina.
E-mail address: pfrances@gl.fcen.uba.ar (P.R. Franceschinis).

https://doi.org/10.1016/j.tecto.2020.228386
Received 22 August 2019; Received in revised form 17 February 2020; Accepted 19 February 2020
Available online 20 February 2020
0040-1951/ © 2020 Elsevier B.V. All rights reserved.

models, this hypothetical ocean was finally closed during the assembly of Pampia in Cambrian times (Escayola et al., 2011; Gosen and Prozzi, 2010). However, paleomagnetic support for the existence of this ocean is still scarce (see Rapalini et al., 2015; Robert et al., 2017; Rapalini, 2018). Two main opposite models have been proposed for the accretion of Pampia to Gondwana: either Pampia experienced a dextral strike-slip displacement from the south of the Kalahari craton up to its current position (see Schwartz and Gromet, 2004; Rapela et al., 2007; Spagnuolo et al., 2012; Franceschinis et al., 2016), during the Cambrian, or the terrane underwent a nearly frontal collision against the Rio de la Plata craton margin in the Ediacaran or Early Cambrian (Kraemer et al., 1995; Rapela et al., 1998; Escayola et al., 2007; among others).

In order to provide further constrains on these tectonic models, we present new paleomagnetic studies in the late Middle to early Late Cambrian Campanario Formation, including preliminary results from the Early Cambrian Puncoviscana metasediments and a nearly coeval intrusive body, exposed in NW Argentina. Together with previous results, these new data allow us to suggest that, in opposition to previously proposed ideas, the strike-slip displaced terrane model cannot be sustained with the available paleomagnetic data, at least for the Middle Cambrian and later times.

2. Geological framework

The Pampia terrane (Ramos et al., 2010 and references therein) has an approximate extension of 1500 km long and 250 km wide. Despite its dimensions and position, surrounded by other cratons and terranes (Fig. 1b), there is still no consensus on its conformation, boundaries, or even on its real existence as an independent terrane. Authors such as Escayola et al. (2011) or Rapela et al. (2015) propose that this terrane does not exist or that it was part of a larger terrane with other blocks such as, the Arequipa-Antofalla block. In this paper, the "conventional" model of Pampia as an independent tectonostratigraphic terrane will be followed (Ramos et al., 2010).

The basement of Pampia is mainly represented by the Puncoviscana Formation *s.l.* that comprises a metasedimentary succession of at least 3000 m thick (Adams et al., 2011) which, in its type locality (Santa Victoria Oeste, province of Salta, Argentina, Turner, 1964), consists in an alternation of greenish sandstone and siltstones with intercalations of tuffaceous and basaltic levels (Escayola et al., 2011). In other sectors, such as around the city of Purmamarca (province of Jujuy, Argentina), the succession shows conspicuous siltstones beds of reddish to greenish colors. The siliciclastic levels have been interpreted as turbiditic (Ježek et al., 1985). The age of the Puncoviscana Formation has been determined as encompassing the Late Ediacarian-Early Cambrian, from different evidence: i) paleontological, such as the presence of the gender *Oldhamia* (Buatois and Mángano, 2003), ii) stratigraphical, as it predates the Tilcárica angular unconformity that separates it from the overlying Middle to Upper Cambrian Meson Group, and iii) radiometric, as presented by Escayola et al. (2011) in intercalated tuffs that yielded an early Cambrian age of 536 ± 5 Ma (U-Pb SHRIMP in zircon) at Quebrada de Humahuaca and 537 ± 1 Ma (U-Pb SHRIMP in zircon) at Puncoviscana locality (Turner, 1964).

In the northern sector of the Eastern Cordillera, the Puncoviscana Formation *s.l.* is intruded by several different plutonic to hypabissal bodies. The Mojotoro intrusive is located on the western slope of the Sierra de Mojotoro near the city of Salta (site MO, Fig. 1c). It consists of a porphyritic granite made up of quartz, plagioclase and potassium feldspar phenocrysts immersed in a fine-grained matrix composed mainly of quartz and plagioclase (Aparicio González et al., 2011). These authors, based on U-Pb LA-ICP-MS dating in zircons, reported a crystallization age of 533 ± 2 Ma for this hypabissal body.

The Mesón Group consists of orthoquarzites, sandstones and conglomerates, with thin intercalations of shales (Turner, 1964). The thickness of the Group varies from tens of meters in the marginal parts of the basin in which it was deposited, up to several hundred meters in the central sectors. Keidel (1943) proposed the subdivision of the Group into three units: Lizoite, Campanario and Chalhualmayoc Formations, a stratigraphic scheme that is still considered valid (Escayola et al., 2011; Spagnuolo et al., 2012; Franceschinis et al., 2016).

The Campanario Formation consists of a basal, green, and a purple upper member. The latter is formed by reddish and purple colored sandstones, characterized by *Skolithus* tubes with interspersed shales. Studies of detrital zircons (U-Pb, LA-ICP-MS) in the Terma de Reyes locality (Jujuy province) carried out by Augustsson et al. (2011) obtained a youngest peak of 514 ± 5 Ma for this unit, thus estimating the maximum age of deposition at 519 Ma. Escayola (unpublished data) has recently obtained a very similar age (519 Ma) as the youngest peak of the age distribution of detrital zircons (U-Pb, LA-ICP-MS) in a sandstone of the Campanario Formation exposed at the Puncoviscana locality, in the extreme north of the province of Salta.

3. Sampling and methodology

3.1. Lower Cambrian units

A preliminary sampling of the Puncoviscana Formation *s.s.* (sites TP1 and TP2, Figs. 1c, 2a, b) was carried out in an outcrop located on national road number 9, near the town of Tilcara (23.7310°S, 65.4662°W). Five oriented hand-samples were collected from the tuffaceous bed, with an approximate thickness of 80 cm, from which Escayola et al. (2011) obtained the 536 ± 5 Ma date. At the same locality, eleven oriented cores were drilled in a reddish sandstone sequence that underlies the tuff, reaching a final sampled thickness for the Puncoviscana Formation of about 3 m.

A small outcrop of the Mojotoro Granite (24.7980° S, 65.3598° W), of porphyritic texture, was sampled on a local road from which nine oriented cores were collected with a portable drill (site MO).

3.2. Middle to Upper Cambrian Unit: Campanario Formation

The Campanario Formation was sampled at three different localities (Fig. 1c), from south to north: Terma de Reyes, Tilcara and El Perchel. Previous studies by Spagnuolo et al. (2008, 2012) in the localities of Matancillas and Parada del Condor (Iruya), respectively, and Franceschinis et al. (2016) near the town of Santa Victoria Oeste also were located in outcrops of this formation, to the north of our study localities (Fig. 1c). With the previous studies, our new localities permit to reach an area of sampling of the Campanario Formation over 200 km long.

Sampling in Terma de Reyes locality (24.1698° S, 65.4820° W) consisted of five sites (R1, R2, R3, R4 and R5). The horizontal distance between the most distant sites (R3 and R5) exceeds 80 m, while the stratigraphic thickness of the sampled sequence is approximately 40 m. A total of 46 oriented cores were collected. The fine-grain sandstones of this formation show an homoclinal attitude at this locality with a NE-SW general strike and bedding dips around 50° to the NW.

At Tilcara (23.7114° S, 65.4545° W), sampling encompassed four sites (T1, T2, T3 and T4). Horizontal distance between the most distant sites (T1 and T4) is approximately 160 m, with a stratigraphic thickness sampled of several tens of meters. A total of 36 oriented cores were collected with a portable drill. As in the previous location, the sequence is basically homoclinal with a NNW-SSE strike and dips 30° to the NW, although, a second-order fold of around one-meter wavelength was observed and sampled (site T3).

At "El Perchel" (23.4918° S, 65.3628° W, Fig. 2c, d), located < 10 km to the north of Tilcara, oriented cores were collected from two sites (EP1 and EP2). They are located some 220 m apart and a difference in stratigraphic position not greater than a couple of tens of meters. With a portable drill, 18 oriented cores were collected. At this locality abundant traces of *Skolithos*, typical of the Campanario Formation, were observed (Fig. 2c). Here, this unit is exposed in an approximately

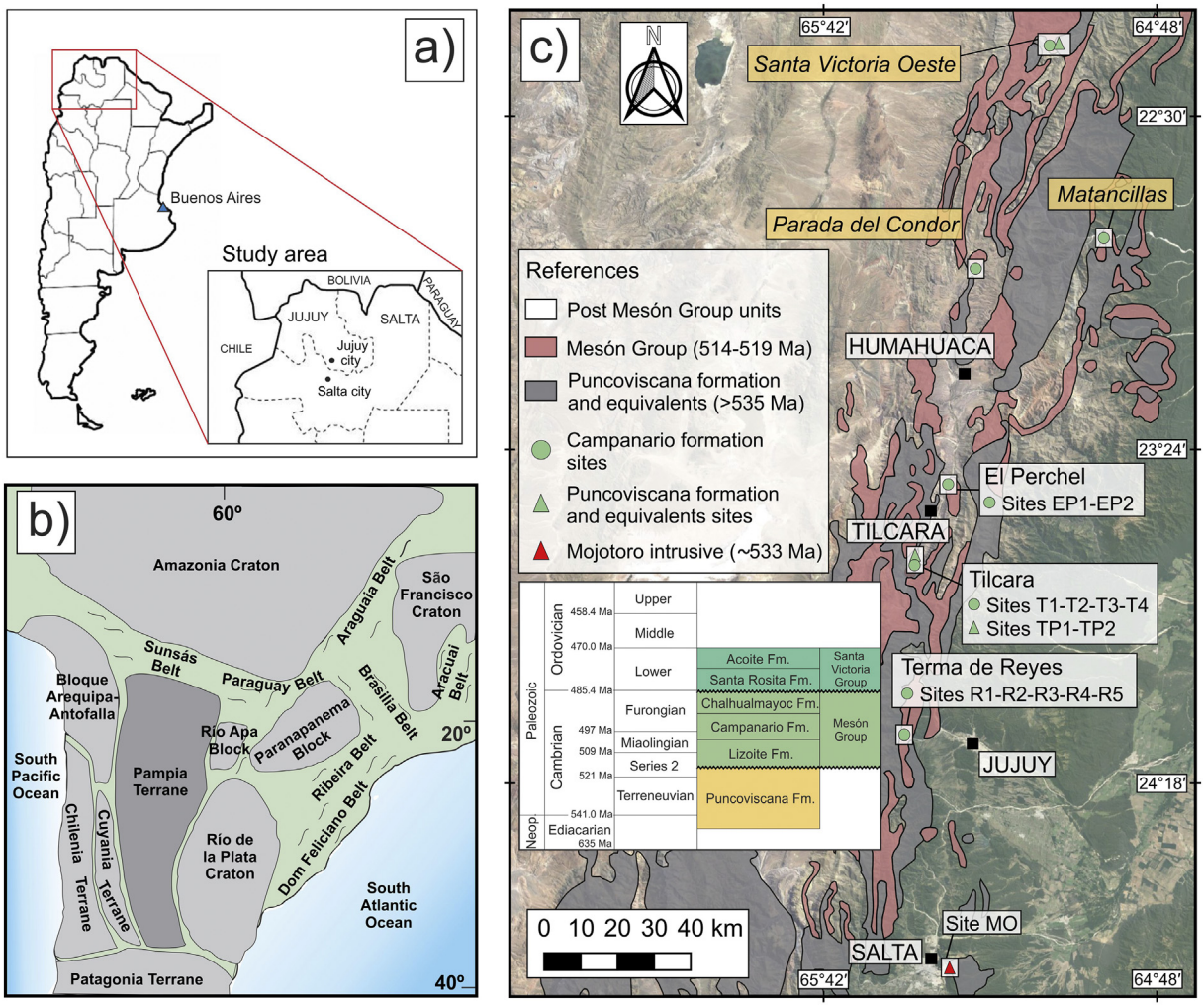


Fig. 1. a) Location of the study area in Salta and Jujuy provinces of Argentina; b) sketch of the Pampia terrane and its relationship with other South American terranes and Proterozoic cratons; c) Sampling locations of the Puncoviscana Formation, the Mojotoro intrusive and Campanario Formation. In italics and yellow, the localities sampled by Spagnuolo et al. (2008, 2012) and Franceschinis et al. (2016). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

homoclinal succession, although some variations in the structural attitude were observed indicating some degree of structural complexity, with a SSW-NNE strike and dips 45° to the NW in EP1 site and a SSE-NNW strike and dips around 60° to the SW for EP2 site.

4. AMS study

4.1. Lower Cambrian units

Bulk susceptibility values of the Puncoviscana Formation samples (sites TP1 and TP2, Fig. 3a, Table 1) show a narrow range between 2.09×10^{-4} and 2.56×10^{-4} [SI] for the tuffs and between 2.94×10^{-4} and 3.79×10^{-4} [SI] for the purple sandstones. The mean anisotropy degree in both sites is virtually identical (Pj: 1.062 and Pj: 1.064 for the tuffs and the reddish arenites, respectively). The shape parameter (T, Jelinek, 1981) is well defined with dominantly oblate fabrics. The layout of the main susceptibility axes show slight differences, particularly in the arrangement of the K3 axes, for which a better grouping is observed around a vertical position for the sedimentary strata when the bedding correction is applied. In any case, an E-W alignment of subhorizontal K1 axes is noticeable, suggesting some tectonic overprint in the AMS of this unit.

The sampled outcrop of the Mojotoro granite (Fig. 3b, Table 1) presents very low bulk magnetic susceptibility values, between 6.40×10^{-5} and 8.40×10^{-5} [SI] and a very low degree of anisotropy (Pj: 1.003). The shape parameter (T) does not show a well-defined fabric. The arrangement of the main susceptibility axes is triaxial, with a K3 axis with a SW orientation and a low inclination angle defining a NW-SE magnetic foliation plane dipping towards the NE.

4.2. Middle to Upper Cambrian Unit: the Campanario Formation

The Campanario Formation in the El Perchel locality (Fig. 4a, Table 1) shows bulk susceptibility values that vary by almost an order of magnitude within the same site, in EP1 between 1.46×10^{-5} and 1.13×10^{-4} [SI] while in EP2 between 2.50×10^{-5} and 7.28×10^{-5} [SI]. Although some specimens show a prolate form of their AMS ellipsoid, the fabric is predominantly oblate. The heterogeneity observed in the values of bulk susceptibility in this locality also appears in the degree of anisotropy that in some specimens exceeds 20%. However, the mean anisotropy degree in the locality is low (Pj: 1.025). The arrangement of the principal axes of anisotropy shows a cluster of subhorizontal K1 (magnetic lineation) in an approximately N-S trend, while axes K2 and K3 are distributed in a subvertical plane of approximate E-W direction. Although there are no major changes when performing the bedding correction, a better definition of the K1 grouping in horizontal position and of the girdle defined by K2 and K3 axes is observed, with several specimens showing subvertical K3. The

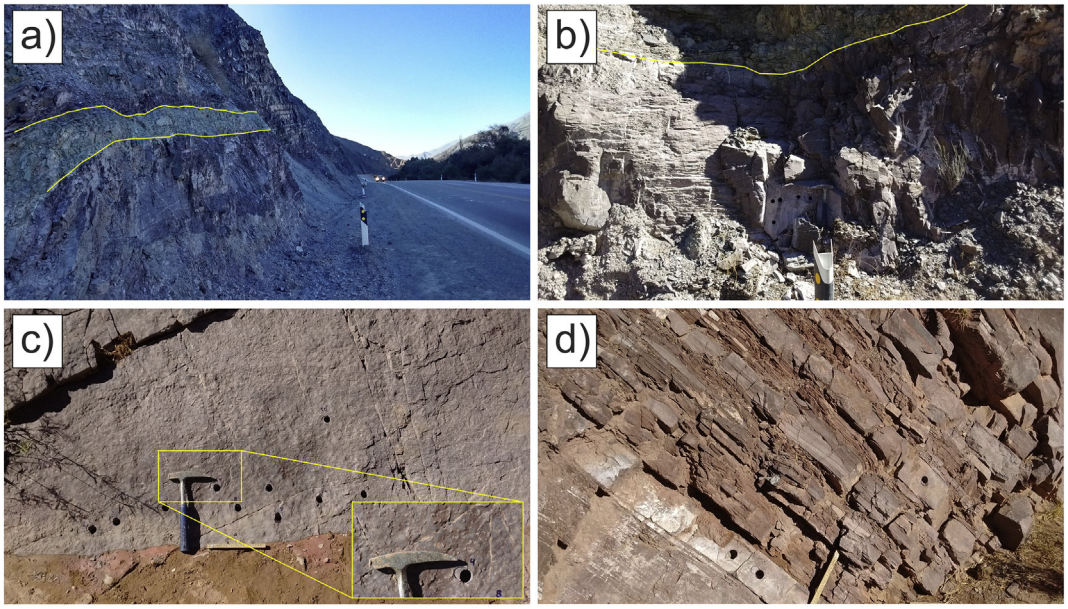


Fig. 2. a) Outcrop of the Puncoviscana Formation. Between the yellow lines, the tuffs sampled in site TP1; b) underlying the previous site, the sampled outcrop of the reddish levels at site TP2; c) Outcrop of the Campanario Formation in the EP1 site (El Perchel locality). In detail, the important amount of traces of Skolithos is shown; d) the outcrops appear as homoclinal sequences (EP2 site). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

AMS pattern observed in this locality can be interpreted as a detrital original fabric with a moderate to strong tectonic overprint (e.g. Pares, 2015). A stretching direction (K1) with approximate N-S to NNE-SSW trend and an E-W compression direction can be inferred.

In Terma de Reyes locality (Fig. 4b, Table 1), two main groups are observed depending on their bulk susceptibility values. On one hand, sites R1, R2 and R3 show values that are between 8.70 × 10^{-5} and 1.65 × 10^{-4} [SI], while sites R4 and R5 reach somewhat higher values that are between 7.43 × 10^{-5} and 2.42 × 10^{-4} [SI] (Fig. 4b). R4 and R5 are found in the highest stratigraphic levels suggesting that these changes in bulk susceptibility may be associated with mineralogical changes along the stratigraphic column. The fabric is mainly oblate and the values of anisotropy degree are low to moderate (average Pj: 1.030). The main axes of susceptibility are well defined with a sub-horizontal eastward directed K3, which defines a plane of foliation of high angle dipping to the W and approximate N-S trend. When the bedding correction is applied, K3 axes become sub-vertical, which indicates that the magnetic foliation plane is close to the bedding plane. The low degree of anisotropy and the poor directional definition of the K1 and K2 axes, added to the near vertical K3 axes after the structural correction, suggest that the magnetic fabric of the Campanario Formation in this locality is pre-tectonic and of detrital origin. The slight angular difference between the magnetic foliation and bedding planes could be due to depositional imbrication. Under this interpretation, paleocurrents at this locality should have been from W to E (present-day coordinates) during deposition of these sediments.

In the Tilcara locality (Fig. 4c, Table 1), the Campanario Formation shows bulk susceptibility values with a narrow range between

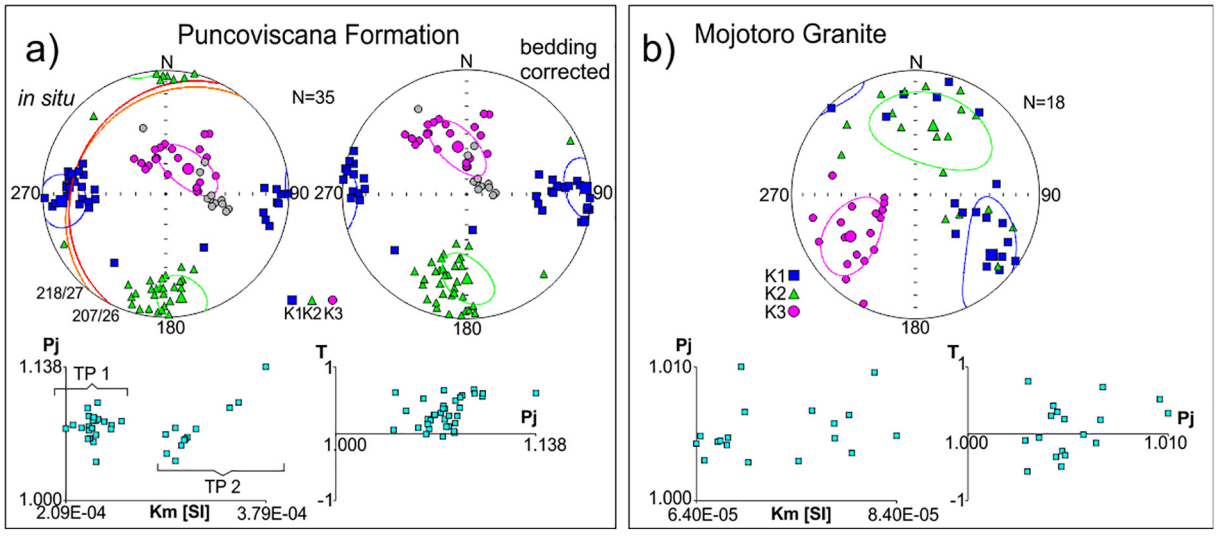


Fig. 3. Stereographic projections for the main axes of the AMS ellipsoid of the Puncoviscana Formation and the Mojotoro granite, and their respective Pj *vs.* Km and T *vs.* Pj plots; a) stereographic projection and diagrams for the Puncoviscana Formation. The grey circles in the stereographic projection indicate the K3 axis of the specimens of the TP2 site (red sandstones); b) idem for the site sampled in the Mojotoro Granite. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Bedding corrected AMS parameters for the sampled units: K1, K2 and K3 that correspond to the maximum, intermediate and minimum axes of the ellipsoid, respectively. Their directions (declination/inclination) are presented with the respective 95% confidence ovals (the maximum and minimum semiaxes are indicated in italics in parentheses). Pj: degree of anisotropy; T: shape parameter (Jelinek, 1981). Cores from site T3 were not considered due to the irregular shape of the specimens for AMS measurements.

Unit	Site	BP (°)	Kmean (μSI)	Pj	T	K1 (°)	K2 (°)	K3 (°)
Mojotoro Granite	MO1	–	72	1.003	0.419	128.3/22.7 (*36.6/17.0*)	14.8/43.5 (*38.9/25.4*)	237.3/37.8 (*30.0/16.4*)
Puncoviscana Formation	TP1	218/27	233	1.062	0.328	88.8/14.4 (*18.6/10.1*)	190.0/37.1 (*15.8/8.4*)	341.5/49.3 (*18.1/12.5*)
	TP2	207/26	319	1.064	0.35	272.3/7.9 (*14.0/7.2*)	180.2/15.1 (*16.1/13.1*)	29.1/72.9 (*16.4/4.3*)
Campanario Formation	EP1	188/44	92	1.034	-0.032	8.0/3.4 (*10.3/4.6*)	273.6/52.0 (*45.8/6.3*)	100.7/37.8 (*45.9/7.6*)
(El Perchel)	EP2	160/57	49	1.049	0.700	14.6/1.8 (*32.1/27.5*)	106.7/48.5 (*34.6/31.3*)	283.1/41.4 (*40.6/5.9*)
Campanario Formation	M2	181/39	86	1.020	0.352	358.3/2.3 (*20.7/5.4*)	267.9/8.7 (*21.0/11.9*)	102.8/81.0 (*12.4/5.5*)
(Santa Victoria Oeste)	M3	189/26	124	1.037	0.786	189.1/2.2 (*29.1/3.8*)	279.5/9.0 (*29.1/6.8*)	85.4/80.7 (*7.1/4.0*)
	M4	174/35	135	1.024	0.712	22.8/8.8 (*12.3/2.6*)	113.3/3.1 (*12.3/8.6*)	222.5/80.7 (*8.8/2.3*)
Campanario Formation	R1	201/48	122	1.022	0.414	347.9/2.6 (*15.0/7.6*)	257.4/9.1 (*15.9/7.5*)	93.6/80.5 (*11.2/9.8*)
(Terma de Reyes)	R2	193/46	104	1.025	0.580	1.3/5.9 (*19.5/4.6*)	270.2/10.0 (*19.5/6.3*)	121.6/78.3 (*6.6/4.6*)
	R3	192/47	122	1.007	-0.012	192.1/7.8 (*24.2/16.2*)	100.4/12.1 (*31.6/23.0*)	314.3/75.6 (*34.1/7.3*)
	R4	195/50	151	1.049	-0.033	336.9/0.2 (*4.8/2.9*)	66.9/1.8 (*3.9/2.8*)	241.1/88.2 (*5.4/2.8*)
	R5	201/46	200	1.057	0.810	207.9/16.8 (*17.8/4.5*)	306.3/25.7 (*17.7/4.8*)	88.3/58.6 (*5.4/4.9*)
Campanario Formation	T1	337/23	216	1.034	0.771	303.7/6.6 (*53.8/5.2*)	213.3/4.1 (*53.9/8.8*)	91.5/82.2 (*13.8/4.2*)
(Tilcara)	T2	342/29	160	1.012	-0.039	315.6/16.1 (*30.3/7.1*)	221.9/12.5 (*30.1/24.0*)	95.8/69.4 (*24.4/9.5*)
	T4	328/29	224	1.016	-0.601	204.9/20.0 (*59.4/13.4*)	305.7/27.2 (*60.0/30.7*)	83.3/55.2 (*48.1/21.0*)

1.01×10^{-4} and 2.78×10^{-4} [SI]. The magnetic fabric also is mainly oblate and the values of anisotropy degree are low (average Pj: 1.019). Site T2 shows lower susceptibility ($< 1.91 \times 10^{-4}$ [SI]) and anisotropy (average Pj: 1.012) values. The main susceptibility axes distribution describes a typical depositional fabric with a K3 axis dipping at high-angle to the SW in geographical coordinates and becoming subvertical when performing the bedding correction. K1 and K2 axes describe a plane of magnetic foliation approximately coincident with the bedding plane. No significant tectonic overprint is observed in this pre-tectonic magnetic fabric. A slight imbrication is apparent on the AMS data, suggesting that at this locality, too, paleocurrents probably were from W to E (present-day coordinates) during deposition of the sampled succession.

5. Paleomagnetic results

5.1. Mojotoro intrusive

After a pilot study to determine the best demagnetization protocol to isolate the magnetic component(s) of this intrusive, stepwise demagnetization by high temperatures was selected as the most suitable method for this purpose. The thermal steps applied were the following: 100, 200, 250, 300, 350, 400, 450, 500, 530 and 560 °C. Most samples presented a relatively stable magnetic memory during demagnetization, with a relatively narrow unblocking temperature range above 530 °C. In most cases univectorial behaviors (Fig. 5a) were observed. The few low-temperature components observed showed high directional dispersion and were excluded from any subsequent interpretation. The characteristic magnetic component was defined up to the maximum demagnetization temperatures by principal component analysis (Kirschvink, 1980) with MAD generally under 10°. The mean characteristic direction for the single site in this porphyritic granite is: n: 11, Dec: 240.5°, Inc.: −10.8°, k: 13, α95: 13.1° (Fig. 5b). Aparicio González et al. (2011) have described this outcrop as a dike (although Toselli and Alonso, 2005 define it as an intrusive and not as a dike) of granitic composition and porphyritic texture indicative of a hypabissal emplacement. Structural studies of the area described a major N-S trending regional anticline (Ruiz Huidobro, 1968; Moya, 1998) and close to the sampled outcrop, the intruded metasediments, assigned to the latest Proterozoic Chachapoyas Formation, as well as younger units affected by the Andean folding have been recognized as dipping between 30° and 40° towards the SW (Ruiz Huidobro, 1968). Our AMS data on these rocks (Fig. 3b) shows a K3 axis with moderate dipping towards the SW. Applying such bedding correction to these data turns mean K3 axis subhorizontal. Since normal magnetic fabrics in dikes tend to show foliation planes subparallel to the dike margins (*e.g.* Hrouda et al., 2016 and references therein), this correction strongly suggests that the sampled site was likely a dike intruded in a subvertical position and tilted during folding of the regional anticline. Considering the AMS (Fig. 3b) results and the regional bedding attitude, a structural correction of Az 147° and 38° dip to the SW was applied to the remanence data (Fig. 5b). Applying this correction, the mean paleomagnetic direction of the site is: Dec: 242.2°, Inc.: −48.7° (Fig. 5b).

5.2. Puncoviscana Formation

The Puncoviscana Formation presented different magnetic behaviors when the pilot study was carried out. Samples from site TP1, corresponding to a tuff, showed that the alternating field method was the most effective to isolate the magnetic components (Fig. 5c). AF demagnetization steps used for this site were: 3, 6, 9, 12, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90 mT. After deletion of a magnetic component with low coercivities (< 30 mT, component "a"), a second component "b" could be defined at fields over 40 mT. This second component trends to the origin of coordinates in the Zijderveld diagrams (Fig. 5c). In those cases in which the samples were demagnetized by high temperatures, component "b" was determined at temperatures below 500 °C. The characteristic component ("b") was defined through principal component analysis (Kirschvink, 1980) with values of MAD < 8°. Samples from TP2, located in the underlying red sandstones, were demagnetized more efficiently by thermal methods, with steps of 100, 200, 250, 300, 350, 400, 450, 500, 530, 560, 590, 620 and 650 °C. The unblocking temperatures were generally higher than 620 °C (Fig. 5c). Component "c" was determined at temperatures lower than 560 °C by principal component analysis with MAD < 14°. In a few cases, a higher temperature component could be isolated at temperatures above 560 °C that did not show a tendency to the origin and directional coherence (Fig. 5c). Samples from both sites collected at Puncoviscana Formation, despite different lithology, magnetic mineralogy (see below) and demagnetizing methods, provided similar characteristic remanence directions (Fig. 5). All of them are located in the NE quadrant with negative (upwards) moderate inclinations in *in situ* coordinates that become shallower when correcting for bedding. The low-temperature/coercivity component "a" at site TP1 has a mean direction *in situ*: Dec: 43.1°, Inc.: −32.1°, k: 34.6, α95: 6.6°, n: 15, whereas when the bedding correction is applied, its mean direction is: Dec: 58.1°, Inc.: −26.1°. On the other hand, component "b" has a mean direction *in situ*: Dec: 53.1°, Inc.: −26.2°, k: 84.3, α95: 3.4°, n: 22 and after bedding correction: Dec:

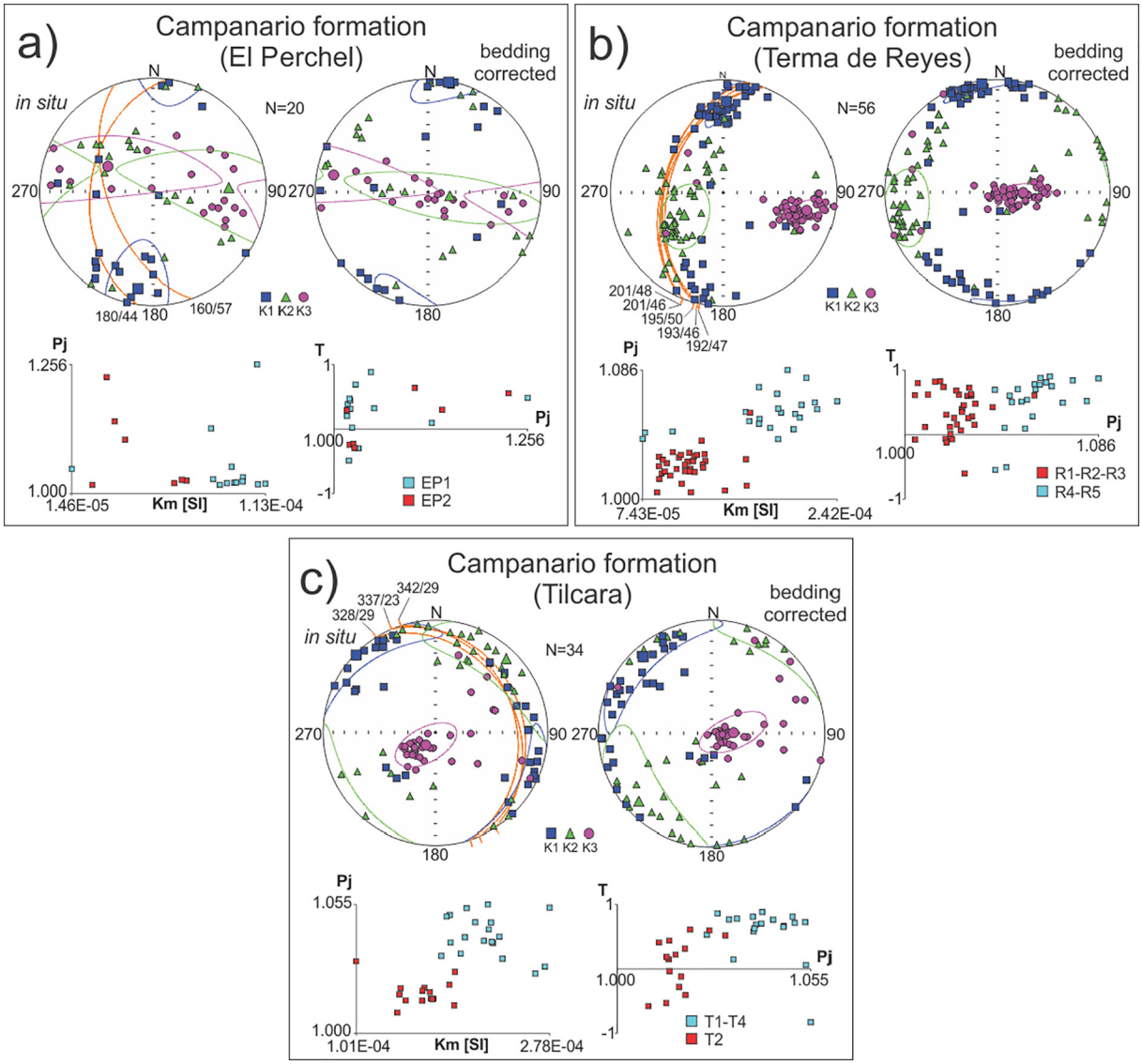


Fig. 4. Stereographic projections for the main axes of the AMS ellipsoid (K1, K2 and K3) of the Campanario Formation in the different localities where it was sampled. a) Stereographic projection and Pj *vs.* Km and T *vs.* Pj diagrams for the sites sampled in El Perchel locality. In red, specimens from the EP2 site; b) idem for Terma de Reyes. In red, samples from sites R1, R2 and R3; in turquoise sites R4 and R5; c) idem for Tilcara. In red, site T2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

63.3°, Inc.: −16.7° (Fig. 5d). Component "a" at site TP2 has a mean direction *in situ*: Dec: 36.2°, Inc.: −43.2°, k: 41.2, α95: 6.3°, n: 14 and after bedding correction is applied: Dec: 56.3°, Inc.: −34.3° (Fig. 5e).

5.3. Campanario Formation

Samples collected at Terma de Reyes locality were demagnetized through high temperatures following the steps: 100, 200, 250, 300, 350, 400, 450, 500, 530, 560, 590, 620, 650 and 670 °C. Components with a lower unblocking temperature, generally between 100 and 500 °C, were found only in a small fraction of the collected samples, meanwhile a high-temperature component "b" with unblocking temperatures over 650 °C (Fig. 6a) was characteristic of most of them. The characteristic component "b" were defined with maximum angular deviation (MAD) < 8° in 46 specimens with an *in situ* mean direction of Dec: 62.7°, Inc.: −0.3°, k: 51, α95: 3.0°, n: 46. When applying the bedding correction, the mean direction becomes Dec: 52.3°; Inc.: 30.9°; k: 45, α95: 3.2° (Fig. 6b). The slight worsening of the statistical parameters is not significant due to the small variation in bedding attitudes at the different sites. As discussed later, however, several indicators suggest that the remanence is pre-tectonic.

Samples from site R3 did not provide positive results to the demagnetization process, presenting a very unstable behavior. Therefore, no remanence directions were computed from this site. The remaining sites show a very high directional consistency (see Table 2), with α95 values under 5°.

At Tilcara, demagnetization of the Campanario Formation also was accomplished by high temperatures, following similar steps. This permitted to isolate two magnetic components. Component "a" with lower unlocking temperatures, between 100 and 450 °C was only found in a few samples, while most of them carried a high-temperature characteristic remanence (component, "b") (Fig. 6c). Demagnetization diagrams show a very narrow and discrete spectrum of unblocking temperatures above 650 °C in which > 50% of the natural remanence is erased. The "a" components with MAD < 13° gave an *in situ* mean direction: Dec: 33.5°, Inc.: −4.7°, k: 43, α95: 11.8°, n: 5, with a significant worsening of the statistical parameters after bedding correction: Dec: 20.8°, Inc.: −30.6°, k: 28, α95: 14.8° (Fig. 6d). Component "b" could be

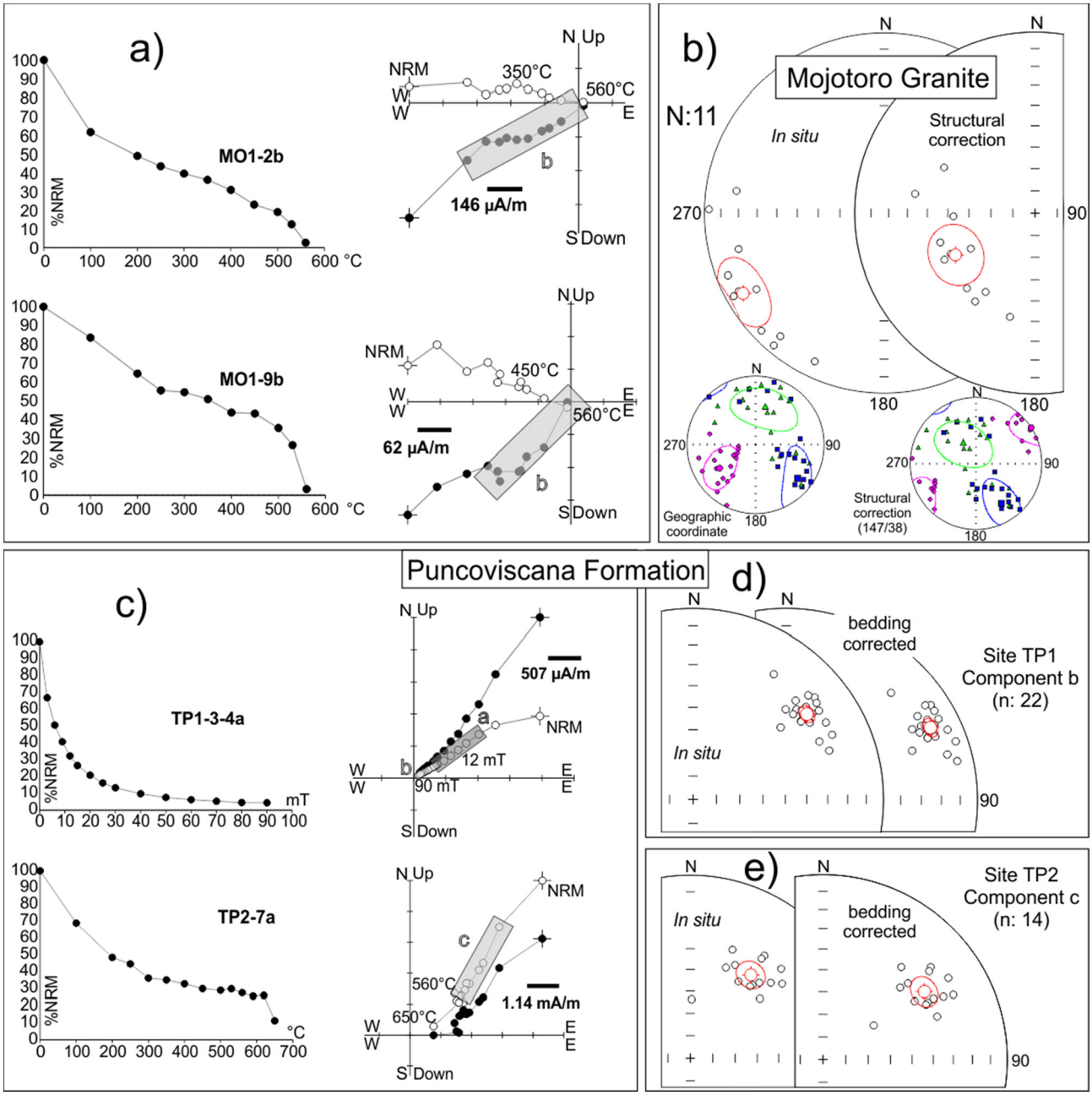
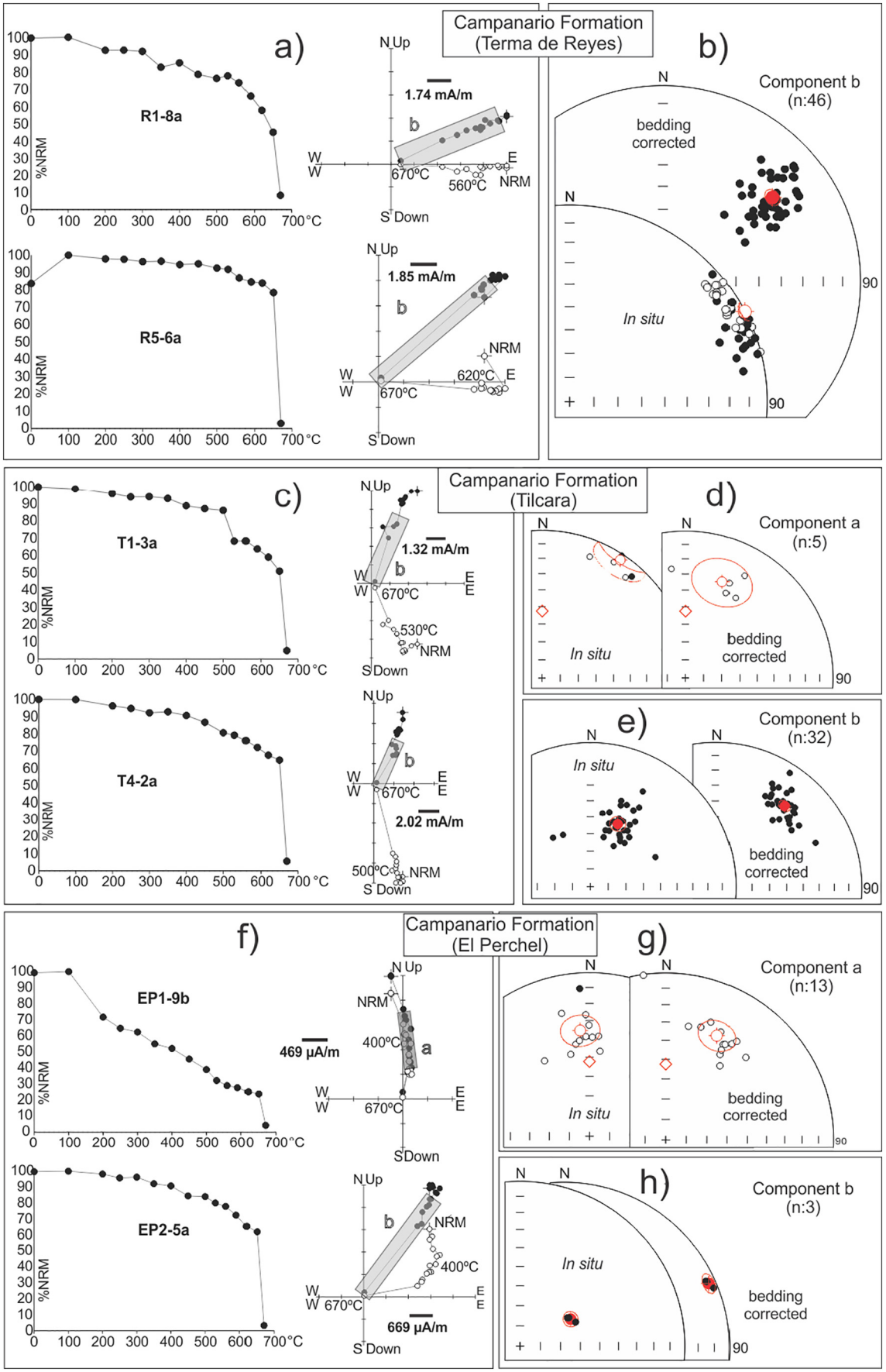


Fig. 5. a) Representative magnetic behavior of the specimens used to determine the mean direction of the Mojotoro granitic dike. Examples of demagnetization by high temperatures are shown. The white (black) symbols in the Zijderveld diagram correspond to the projection in the vertical (horizontal) plane; b) directions of the characteristic remanence of specimens from the Mojotoro granitic dike in geographical coordinates and after structural correction (see text on this issue). Stereographic projections for the main axes of the AMS ellipsoid of the Mojotoro intrusive also are shown. Note the arrangement *in situ* and after bedding correction (147°/38°); c) idem a) for Puncoviscana Formation (TP1 corresponds to tuffs and TP2 to red sandstones). Demagnetization examples are shown by alternating fields and high temperatures; d) specimens directions of component "b" *in situ* and after bedding correction; e) idem for specimens directions of the characteristic remanence (component "c") of the Puncoviscana Formation. In light grey shadow components "b" and "c" and in dark grey shadow component "a" that were isolated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

defined with MAD lower than 9°, except for two specimens. The specimens mean direction *in situ* is Dec: 23.6°, Inc.: 51.1°, k: 36, α95: 4.3°, n: 32 while after bedding correction is Dec: 39.8°, Inc.: 27.5°, k: 69, α95: 3.1° (Fig. 6e). The improvement in the grouping of component "b" is statistically significant. At this locality, as already mentioned, site T3 was located on a highly dipping limb of a second-order fold. Applying a fold test, a positive result is obtained with a maximum τ1 value at 110% unfolding (Fig. 7, see Koymans et al., 2016; Tauxe and Watson, 1994). This indicates that the characteristic magnetization of the Campanario Formation at Tilcara locality is pre-tectonic. Mean site data is presented in Table 2. Unfortunately, T3 mean direction was obtained from just two independent cores, therefore no significant statistical parameters could be computed for it. The remaining sites show moderate to high directional consistency (α95 < 9°).

Samples of the Campanario Formation at El Perchel locality were demagnetized by high temperatures following the steps: 100, 200, 250, 300, 350, 400, 450, 500, 530, 560, 590, 620, 650 and 670 °C. It was possible to isolate two magnetic components by principal component analysis. An "a" component of low to moderate unblocking temperatures, generally isolated between 100 and 580 °C with MAD under 13°,



(caption on next page)

Fig. 6. a) Representative magnetic behavior of specimens used to determine the mean direction of the remanent magnetization of the Campanario Formation in Terma de Reyes locality. Examples of demagnetization by high temperatures are shown. The open (full) symbols in the Zijderveld diagram correspond to the projection in the vertical (horizontal) plane; b) specimen directions of high-temperature component "b" *in situ* and after bedding correction; c) idem a) for Campanario Formation in Tilcara locality; d) specimen directions of low-temperature component "a" *in situ* and after bedding correction; e) idem b) in Tilcara locality; f) idem a) for Campanario Formation in El Perchel locality; g) idem d); h) idem e). Red diamond: axial dipole field at the study locality. In light grey shadow components "b" and in dark grey shadow component "a" that were isolated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and a high temperature component "b" determined at temperatures higher than 580 °C (MAD < 10°) (Fig. 6f). A moderate directional consistency was obtained for component "a", meanwhile component "b" was properly isolated in only three cores from site EP-2 that provided consistent directions. The mean direction of component "a": Dec: 355.0°; Inc.: −35.4°, k: 23, α95: 8.8°, n: 13, does not experience any significant statistical change when the bedding correction is applied: Dec: 26.0°; Inc.: −31.0°; k: 27, α95: 8.1° (Fig. 6g). The mean *in situ* direction obtained for component "b": Dec: 66.1°, Inc.: 4.7°, α95: 3.8°, n: 3 turns into Dec: 62.0°, Inc.: 61.5° when performing the bedding correction (Fig. 6h).

6. Directional analysis

Fig. 8 shows the mean directions of the characteristic remanence obtained in the Lower Cambrian units of the Pampia terrane. These are the Mojotoro Granite and the Puncoviscana Formation.

The mean direction of the Mojotoro Granite (*ca.* 533 Ma) is subhorizontal and points to the SW in *in situ* coordinates and becomes of moderate negative inclination (upwards) in the SW direction after the structural correction is applied. The *in situ* direction does not coincide with any expected post-Cambrian direction for Gondwana and South America. However, after bedding correction it appears as nearly antipodal to those previously obtained for the Middle to Late Cambrian Campanario Formation (Spagnuolo et al., 2008, 2012; Franceschinis et al., 2016). On the other hand, the Puncoviscana Formation is represented by two mean directions in the NE quadrant with negative inclinations, corresponding respectively to TP-1 (tuff) and TP-2 (red sandstones) sites (Fig. 8), both before and after bedding correction.

In the case of the Campanario Formation, the low-temperature components showed coherent directions in El Perchel and Tilcara localities (Fig. 9). In the latter, however, only five samples carry it. The mean direction of this component in Tilcara becomes closer to the Geocentric Axial Dipole (GAD) direction after bedding correction is applied (Fig. 6g). In El Perchel, however, the mean direction *in situ* is closer to the GAD direction (Fig. 6h). Fig. 9 shows that there seems to be no clear relationship between the low temperature component at El Perchel and Tilcara with any of the remanent components isolated in the Puncoviscana samples.

The mean directions of the characteristic component in the Campanario Formation are presented in Fig. 10 (*in situ* and after bedding correction). They are characterized by small circles of confidence and positive to subhorizontal "*in situ*" inclinations in the NE quadrant. In Fig. 10c, d, overall mean remanence directions are presented together with those obtained at other localities by previous studies (Spagnuolo et al., 2012; Franceschinis et al., 2016). After bedding correction, with the only exception of the El Perchel, all mean directions show similar inclination values with some scatter in declinations.

Undoubtedly, the El Perchel mean direction has not averaged out the paleosecular variation due to its very low number of samples. Whether or not this has been averaged out in the other localities may be subject to discussion. Conservatively, site mean directions were computed for the Campanario Formation. Averaging site mean directions from the different localities should avoid this problem. The "*in situ*" and "bedding corrected" site mean directions from the study localities of the Campanario Formation (including those from Santa Victoria Oeste (Franceschinis et al., 2016) are shown in Fig. 10 c) and d). An overall mean direction from eleven sites (119 specimens) is presented in

Table 2
Mean directions of the Puncoviscana and Campanario Formations *in situ,* after bedding correction, and after application of experimental inclination shallowing corrections. n: number of specimens, N: number of sites, Dec: declination, Inc.: inclination, α95: confidence angle, w/d: without data.

Unit	Site	n (N)	Strike/Dip (°)	*In situ*			Bedding/Structural corrected			IRM corrected		
				Dec (°)	Inc (°)	α95 (°)	Dec (°)	Inc (°)	α95 (°)	Dec	Inc	α95 (°)
Mojotoro Granite	MO1	11	147/38	240.5	−10.8	13.1	242.2	−48.7	13.1	–	–	–
Puncoviscana Fm.	TP1	22	218/27	53.1	−26.2	3.4	63.3	−16.7	3.4	63.8	−53.3	5.2
	TP2	36	207/26	36.2	−43.2	4.4	56.3	−34.3	4.1	56.4	−36.7	6.3
Campanario Fm. (El Perchel)	EP2	3	160/57	66.1	4.7	–	62.0	61.5	–	62.0	61.5	–
Campanario Fm. (Terma de Reyes)	R1	12	201/48	69.1	2.2	3.5	56.1	35.3	3.5	56.1	36.5	3.5
	R2	12	193/46	67.5	7.9	4.5	51.7	42.5	4.5	51.7	45.0	4.4
	R4	8	195/50	56.1	−4.6	4.5	47.8	26.8	4.5	47.8	28.8	4.6
	R5	14	201/46	56.7	−7.0	4.4	52.3	19.3	4.4	52.3	21.7	4.5
	R	46		62.7	−0.3	3.0	52.3	30.9	3.2	52.3	32.9	3.2
	R	(4)		62.4	−0.4	11.0	51.9	31.0	12.0	51.9	33.0	11.9
Campanario Fm. (Tilcara)	T1	8	337/23	30.6	39.2	8.7	37.8	19.8	8.7	37.7	24.2	8.9
	T2	14	342/29	23.6	52.2	3.3	40.2	29.6	3.3	40.1	34.3	3.3
	T3	2	358/71	327.3	56.0	–	53.1	32.6	–	53.1	37.6	–
	T4	8	328/29	26.0	56.4	4.4	38.2	30.0	4.4	38.2	33.9	4.4
	T	32		23.6	51.1	4.3	39.8	27.5	3.1	39.8	31.9	3.1
	T	(4)		14.8	53.6	22.5	42.1	28.1	9.6	42.0	32.7	9.5
Campanario Fm. (Santa Victoria Oeste)	M3	32	189/26	65.9	−2.0	3.6	63.6	19.6	3.6	63.7	24.0	4.0
	M4	6	174/35	62.7	9.7	17.4	55.4	41.7	17.4	56.2	47.9	17.0
	M	38		65.5	−0.2	4.0	62.6	23.0	4.5	62.8	27.6	4.8
	M	(2)		65.7	−1.1	–	60.0	30.7	–	60.5	36.0	–
Campanario Fm.	C2 (EP2-R-T-M)	(11)		52.0	20.8	21.2	50.2	32.8	8.3	50.2	36.2	7.9
Campanario Fm. (Matancillas + Iruya)	C	59	w/d	92.0	53.9	9.5	81.5	37.7	4.3	81.5	37.7	4.3
	C	(8)	w/d	80.8	69.5	41.8	78.3	36.5	10.8	78.3	36.5	10.8

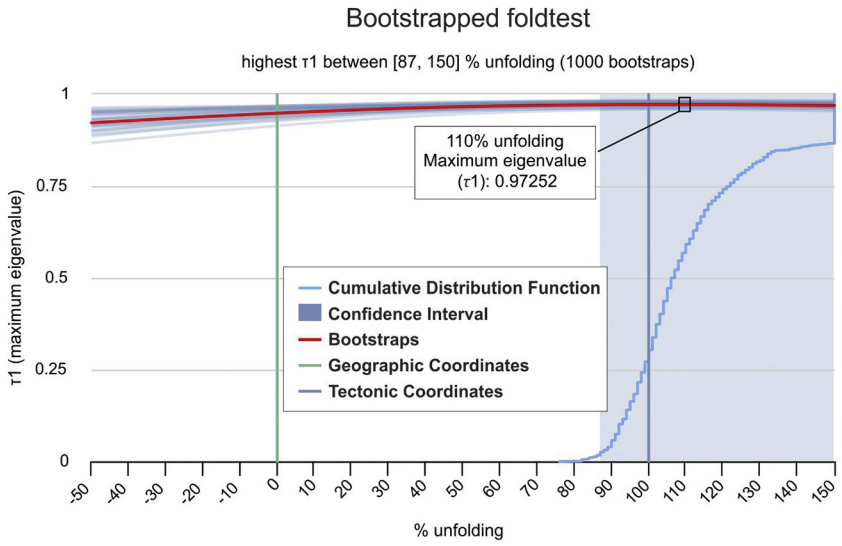


Fig. 7. Fold test carried out in Tilcara locality demonstrating its pre-tectonic character. Image downloaded and modified from de Paleomagnetism.org (Koymans et al., 2016).

Table 2. Only those remanence directions obtained from this formation at the Matancillas and Iruya localities (Spagnuolo et al., 2008, 2012) have been left out of the Campanario grand mean as they clearly show a significant declination anomaly with respect to the other localities (see discussion further on).

7. Quantification of the inclination shallowing effect

Remanence directions in sedimentary rocks may be affected by what is called "inclination shallowing" (*e.g.* Kodama, 2012, and references therein). The experimental way to determine it in a sedimentary sequence and to quantify it in order to correct it is through the analysis of an anisotropic acquisition of an artificial remanence (eg, Hodych and Buchan, 1994; Tan and Kodama, 2002). Following the model proposed by Hodych and Buchan (1994), experiments were carried out imparting an IRM at a known angle with respect to bedding in three samples of the Puncoviscana Formation (sites TP1 and TP2, Table 3) and in 6 samples of the Campanario Formation from different sampling locations: Tilcara (3 samples, from sites T1, T2 and T4, respectively) and Terma de Reyes (2 samples, from sites R1 and R5, respectively, see Table 3). The method consists in artificially magnetizing a sample with a magnetic pulse in a known direction that is close to 45° with respect to the original bedding of the specimen. If there is "inclination shallowing" of the remanence in the studied samples, lower inclination values for the artificial remanence will be obtained than those of the applied field, so that a correction factor can be calculated. For the study of site TP1 in the Puncoviscana Formation, in which ferrimagnetic minerals are the main carriers of the remanence (see next section), fields of 29, 61, 120, 300, 600 and 1000 mT were applied, while in the specimens belonging to the TP2 site and the Campanario Formation, in which antiferromagnetic minerals are the main carriers of remanence (see next section), fields of 600-760-900-1000 mT were applied.

This study allows quantifying the deflection of the remanence vector by compaction, for which correction factors (F) were obtained, from the equation of King (1955):

$$F = \tan Iirm / \tan Ia$$

being *Iirm* the inclination of the artificial remanence and *Ia* the inclination of the applied magnetic field. The correction applied to the inclination of the natural remanence is formulated as follows:

$$Ic = \operatorname{atan}(\tan Io / F)$$

where *Ic* is the corrected inclination and *Io* is the inclination of the characteristic remanence.

The laboratory tests in the Puncoviscana Formation yielded significantly different results for the TP1 site (tuff), with a strong deformation of the magnetization vector towards the bedding layer (average correction factor F: 0.21), in comparison with TP2 (red sandstone, F: 0.90). This is reflected in a very large variation in the mean inclination of the TP-1 site if the correction factor is applied

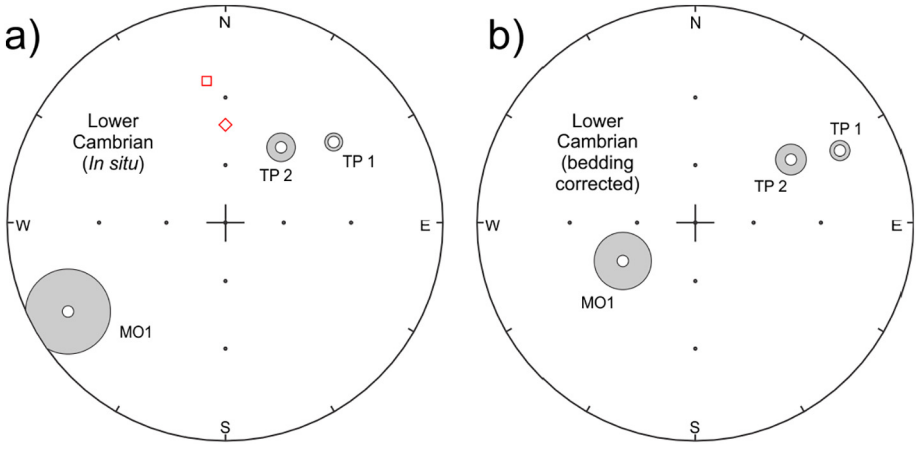


Fig. 8. Mean characteristic remanence directions from the Mojotoro granitic dike (MO1) and the Puncoviscana Formation (sites TP1 and TP2). a) Mean directions *in situ*; b) Mean directions after bedding correction. Red square: present geomagnetic field at the study area; red diamond: direction of the geocentric axial dipole field direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

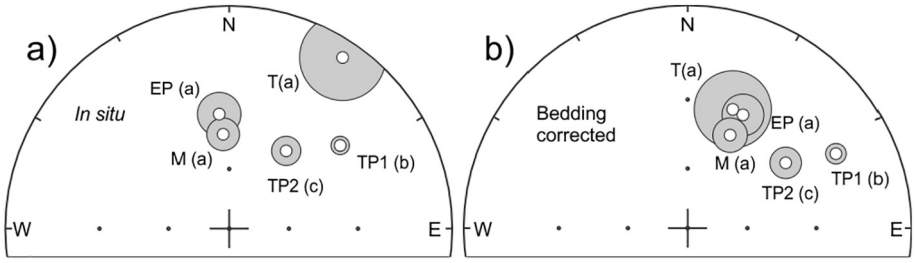


Fig. 9. Mean characteristic remanence directions of low-temperature components (a) that could be isolated in the locations where the Campanario formation was studied. They are compared with the main components (b) and (c) isolated at the TP1 and TP2 sites, respectively, in the Puncoviscana formation. Note that there is no overlapping of components "a" with the main components "b" and "c" neither *in situ* nor after the paleohorizontal correction. M: Santa Victoria Oeste locality (Franceschinis et al., 2016).

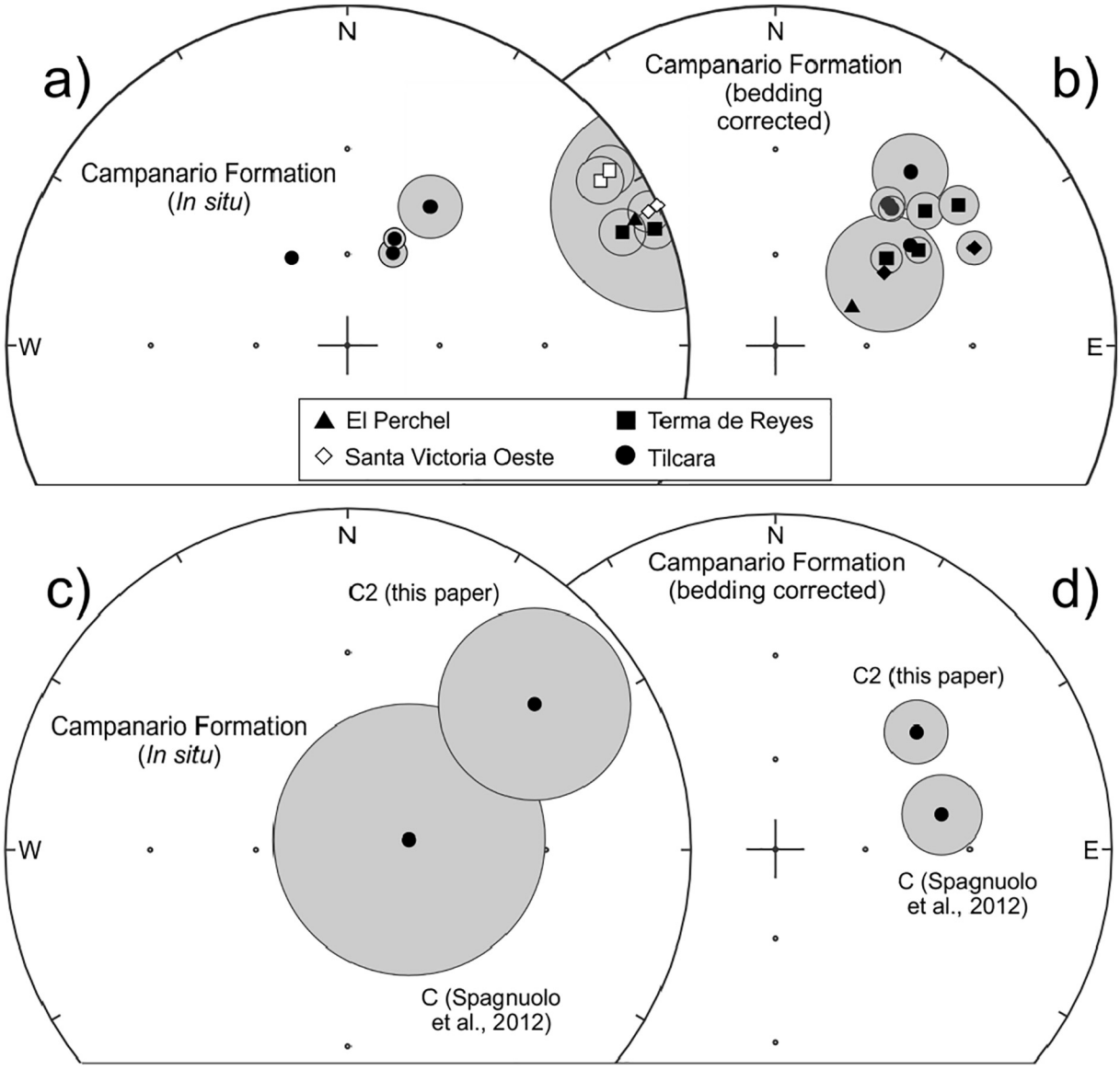


Fig. 10. Mean characteristic remanence directions of each Campanario Formation site. a) Mean directions *in situ*; b) Mean directions after bedding corrections. Site mean directions from Santa Victoria Oeste locality (Franceschinis et al., 2016) have been included; c) Overall *in situ* mean direction of the Campanario Formation sites presented in a) and that from Spagnuolo et al. (2012); d) idem c) after bedding corrections.

(mean inclination changes from 16.7° to 53.3°), while in the other site the variation is almost negligible (34.3° to 36.7°).

Fig. 11a shows the change in the mean directions per site in the rocks studied in the Puncoviscana Formation. It can be observed that the mean directions of both sites become closer after the correction by compaction, although their circles of confidence do not overlap. This may be due to the fact that none of the sites average out the paleosecular variation. It is also possible that the "inclination shallowing" correction values are overestimated in TP-1 and/or underestimated in TP-2.

In the case of the Campanario Formation, *F* values obtained for each site were applied to each specimen characteristic magnetization of such site (Table 3). For Sites R-2 and R-4 for which no reliable results were obtained in the IRM test an average *F* value of those obtained for site R-1 and R-5 was used. The corrected directions showed minor changes (between 2° and 4°) in their inclinations (Fig. 11b, Table 2). The

Table 3
Results obtained from the experiment carried out by isothermal remanent magnetization to quantify the amount of "inclination shallowing". Inclination is respect to the bedding layer in each specimen. The successive fields applied during the experiment are indicated. F: correction parameter. n: number of IRM steps applied. α95: fisherian confidence angle for the mean of IRM directions.

Unit	Sample	Applied field Inc (°)	IRM Inc (°)	Steps (mT)	α95	n	F
Puncoviscana	TP1-5-3	52.7	18.4	29-61-120-300-600-1000	1.1	6	0.25
	TP1-3-1A	42.8	9.4	29-61-120-300-600-1000	1.5	6	0.18
	TP2-5	48.6	45.9	600-760-900-1000	1.0	4	0.90
Campanario	T1-6A	59.2	49.4	600-760-900-1000	1.9	4	0.69
	T1-11A	63.8	62.2	600-760-900-1000	1.4	4	0.93
	T2-9A	61.0	56.3	600-760-900-1000	1.0	4	0.83
	T4-9A	41.8	37.6	600-760-900-1000	3.2	4	0.86
	R1-10A	47.9	46.6	600-760-900-1000	0.6	4	0.95
	R5-9B	46.0	42.3	600-760-900-1000	0.6	4	0.88

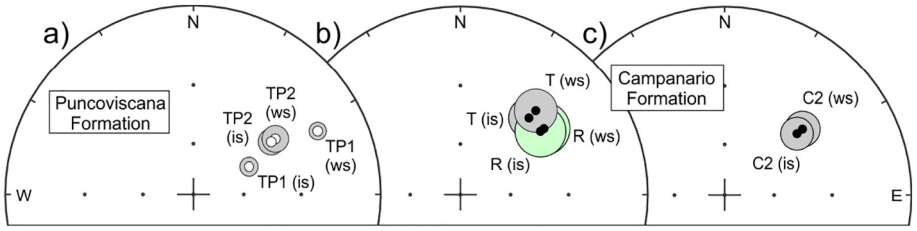


Fig. 11. a) Mean characteristic remanence directions of the Puncoviscana Formation with the correction for inclination shallowing; b) idem a) for the sites studied in Campanario Formation at Tilcara (T) and Terma de Reyes (R) localities; c) Overall site mean directions C2 before and after correction for inclination shallowing. ws: without inclination shallowing correction, is: with inclination shallowing correction. Values are presented in Table 2.

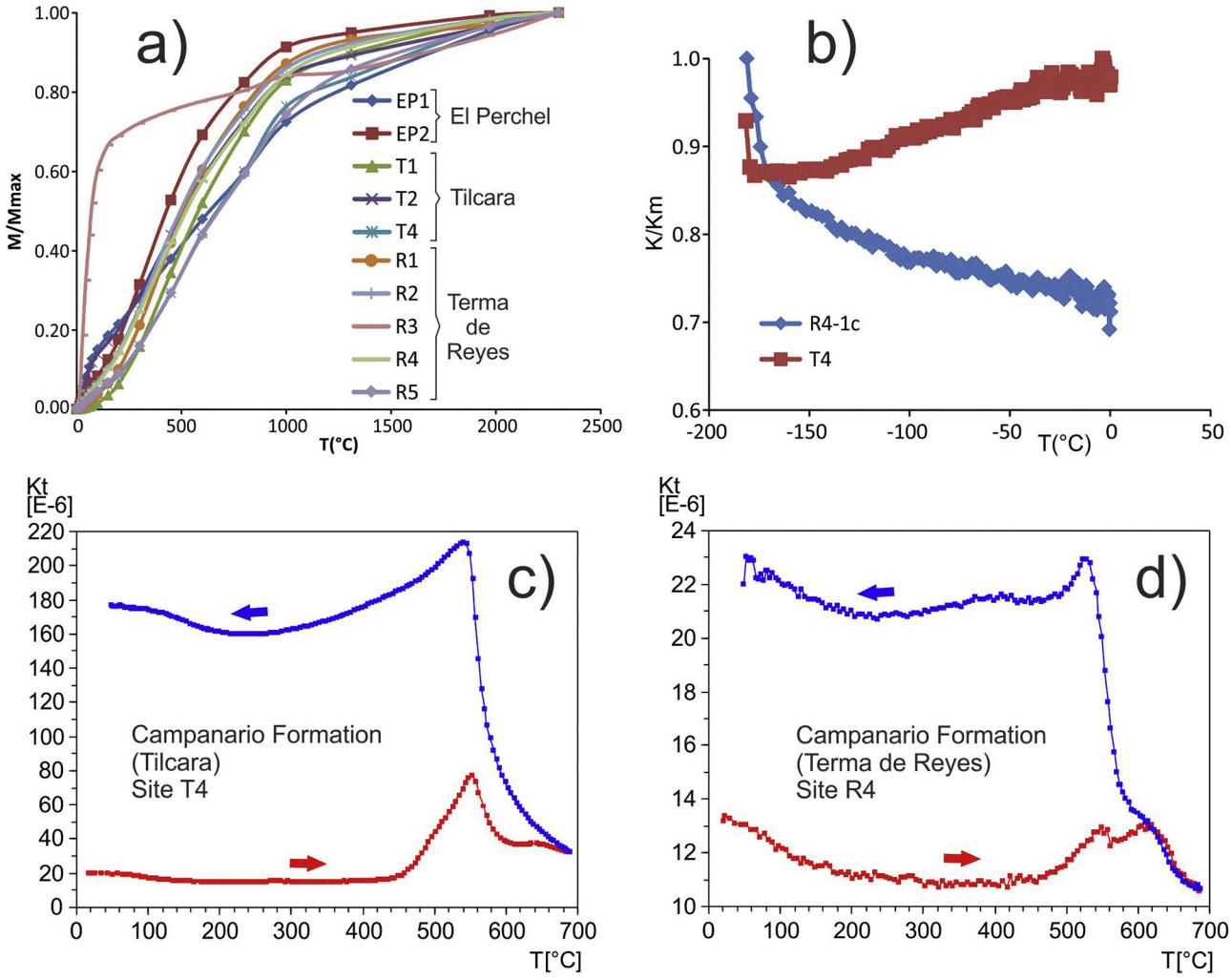


Fig. 12. a) Normalized isothermal remanent magnetization (IRM) acquisition curves for the Campanario Formation. Most samples show curves typical of a dominant/exclusive antiferromagnetic phase, with the exception of R3 in which the ferrimagnetic contribution is dominant; b) Knorm *vs.* low temperature (T) curves; c) and d) high temperature Knorm *vs* high temperature curves. See description in the text.

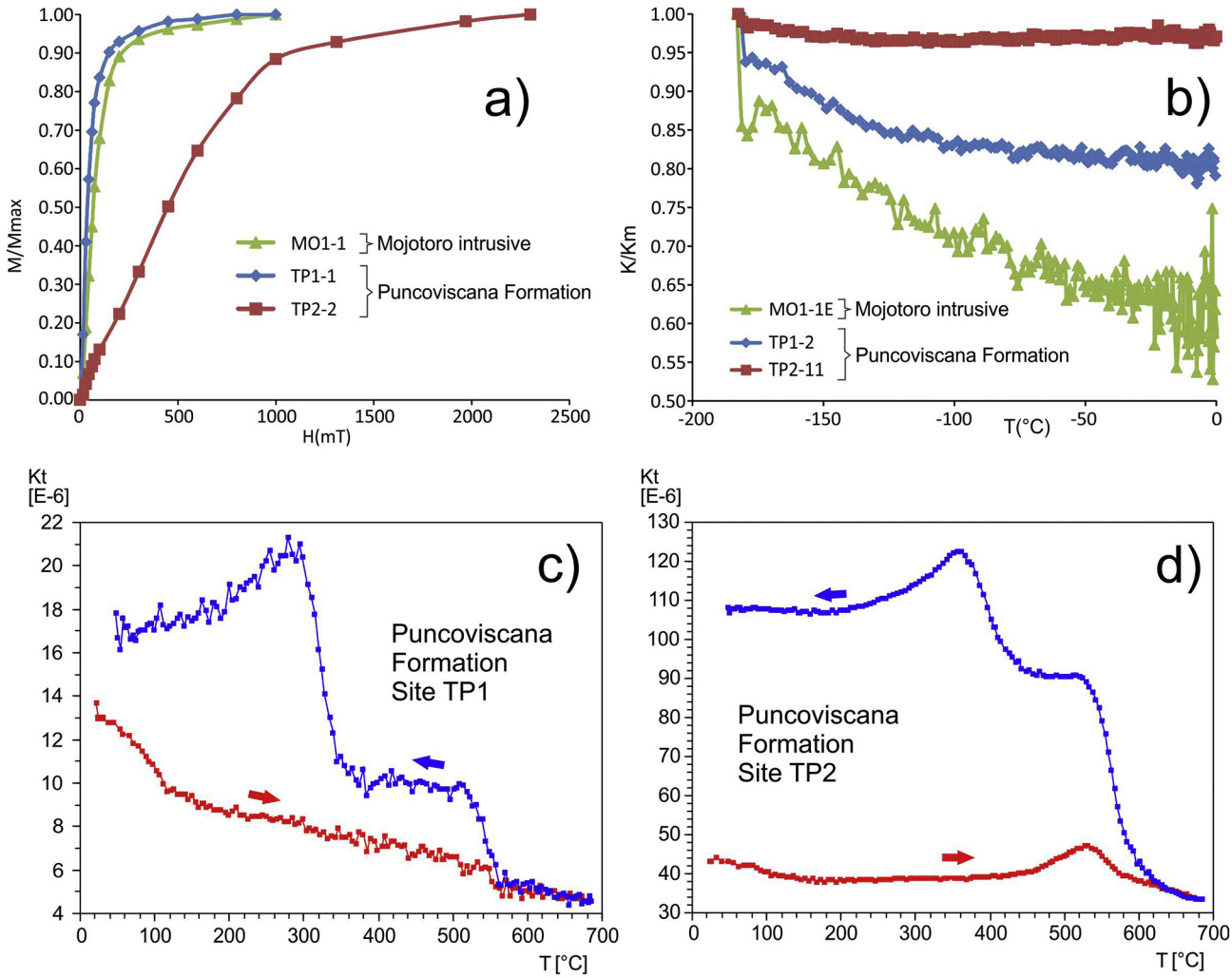


Fig. 13. a) Normalized curves for the acquisition of isothermal remanent magnetization (IRM) for Lower Cambrian studied units of the Pampia terrane. The sites show typical behavior of ferrimagnetic minerals with the exception of site TP2, which is dominated by an antiferromagnetic signal; b) Knorm *vs.* low temperature standard curves temperature (T). Note the hyperbola caused by paramagnetic minerals at sites TP1 and TP2; c) and d) high temperature curves.

modified mean inclination in Tilcara goes from 27.5° to 31.9° and from 30.9° to 32.9° in Terma de Reyes (see Table 2). These values are consistent with previous analysis by Spagnuolo et al. (2008) and Franceschinis et al. (2016) in the Campanario Formation at other localities, where they respectively found, no significant or little inclination shallowing. Site mean directions were recomputed with the specimen-corrected ChRM and are presented in Table 2. The Campanario site grand mean (that only excludes Matancillas-Iruya) also show a very minor change in its direction (Dec: 50.2°, Inc.: 32.8°, α95: 8.3° *vs.* Dec: 50.2°, Inc.: 36.2°, α95: 7.9°, Fig. 11c).

8. Rock magnetism

The IRM acquisition curves for representative samples of the Campanario Formation show typical behavior of rocks whose remanence is dominated by antiferromagnetic minerals, with magnetizations that saturate above 2 T. The concave character of the curves at low fields indicates lack of ferrimagnetic minerals in these samples. The only exceptions to this observation are samples from EP1 at El Perchel and from R3 at Terma de Reyes, with roughly 15% and 70% of the isothermal remanence acquired by a ferrimagnetic fraction, respectively (Fig. 12a). It is noteworthy that no coherent ChRM could be isolated in those sites.

On the other hand, bulk susceptibility curves *vs.* low temperatures (Fig. 12b) show hyperbolas at site R4 (Terma de Reyes), which corresponds to typical susceptibility behaviors dominated by paramagnetic minerals. T4 (Tilcara) shows an increase in susceptibility values when temperature increases, which is usually a diagnosis of the presence of superparamagnetic minerals (SP, Worm and Jackson, 1999).

Bulk susceptibility *vs.* high temperatures curves (Fig. 12c, d) show irreversible cycles in all cases, with formation of magnetite from 400 to 500 °C, as it can be observed by a gradual increase in susceptibility at those temperatures. This mineral neoformation is more evident in the cooling curves, which in some cases show a five-fold increase in bulk susceptibility. The heating curves also show a flat or low slope curve up to temperatures close to 700 °C (Fig. 12c) and a drop in susceptibility between 600 and 700 °C (Fig. 12d), consistent with the presence of an antiferromagnetic phase with Curie temperature close to 700 °C and suggestive of the presence of hematite in these samples.

Isothermal remanent magnetization (IRM) acquisition curves show typical behaviors of ferrimagnetic minerals, with saturating fields under 1 T (Fig. 13a) in the Mojotoro Granite (Site MO1) and in the tuffaceous level of the Puncoviscana Formation (TP1). TP2 site (purple sandstones) shows prevailing antiferromagnetic phases, as saturation was not reached under 2 T (Fig. 13a). A small increase in remanence at low fields, suggests, however, that there is a subsidiary fraction of ferrimagnetic minerals at this site.

Curves of bulk susceptibility *vs.* low temperatures are in the form of

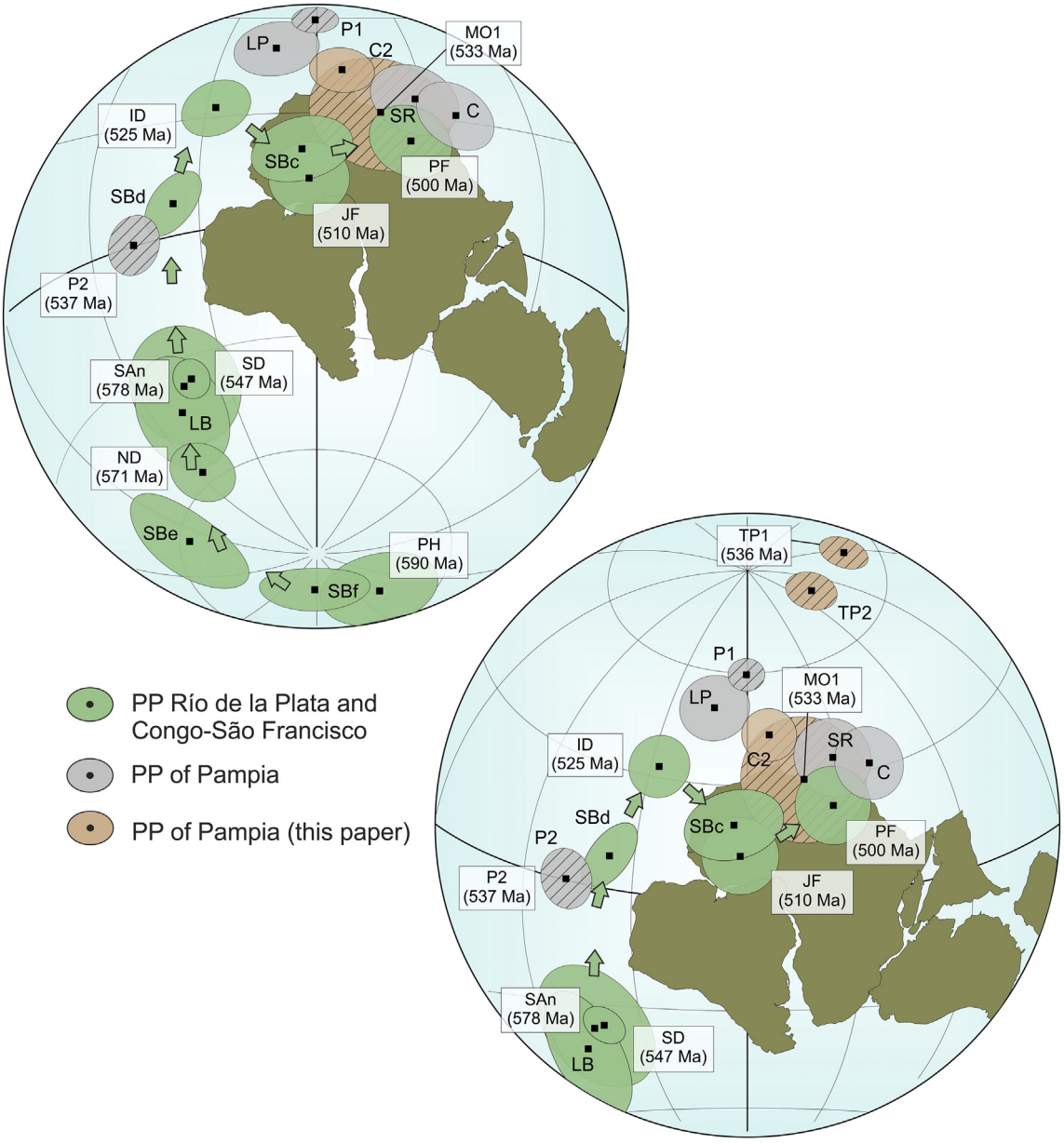


Fig. 14. Paleomagnetic poles of Pampia terrane, Río de la Plata and Congo-São Francisco cratons, for the Early Ediacaran-Ordovician interval (Schmidt projection). The poles appear with their respective 95% confidence circles. The poles were rotated to African coordinates according to Torsvik et al. (2012). The numbers in parentheses indicate the most likely age of the pole, PH: Playa Hermosa Formation; SBf: Villa Monica Formation; SBe: Cerro Largo Formation; LB: Los Barrientos; San: Sierra de Animas; SBd: Olavarría Formation; SBc: Cerro Negro Formation; ND: Nola Dykes; SD: Sinyai Dolerite; ID: Itabiana Dykes; JF: Juiz de Fora Complex; PF: Piquete Formation; P1, P2, TP1, TP2: Puncoviscana Formation; C: Campanario Formation at Matancillas-Iruya; C2: Campanario mean pole (Terma de Reyes, Tilcara, El Perchel and Santa Victoria localities), MO: Mojotoro Granite VGP, SR: Santa Rosita Formation, LP: La Pedrera Formation. VGPs are indicated with a striped plot.

hyperbolas in the Puncoviscana Formation (sites TP1 and TP2, Fig. 13b) indicating a predominant contribution of paramagnetic minerals. High temperature K-T curves show irreversible cycles in the Puncoviscana Formation (Fig. 13c, d). In both sites, notorious increases in susceptibility are observed in the cooling curves at ~560 °C indicating magnetite formation, and around 300 °C to 360 °C suggesting maghemite or pyrrhotite formation during the experimental procedure.

9. Paleogeography of Pampia in the Cambrian and Ordovician

A paleomagnetic pole was computed for the Campanario Fm. on a specimen basis, for each locality. This follows the criteria adopted by Franceschinis et al. (2016) that the paleosecular variation might have been averaged out at each locality, with the exception of El Perchel. However, since only a few sites were actually sampled at each locality, a more conservative approach was chosen an a single paleomagnetic pole was computed for the Campanario Formation, by averaging the corresponding VGP for each site mean direction computed at Terma de Reyes, Tilcara, El Perchel and Santa Victoria Oeste (23.6° S, 346.5° W, A95: 7.0°, N:11). Remanence directions of this formation computed by Spagnuolo et al. (2008, 2012) at Matancillas and Iruya localities have been excluded due to the high likelihood of being affected by tectonic rotations.

The Campanario mean pole computed in that way attain a maximum quality factor “Q” of 7 in Van der Voo's qualification scheme (“Q”, Van der Voo, 1990). T passes a local fold test but T3 site is represented by just two specimens.

On the other hand, the VGPs belonging to the Puncoviscana

Table 4
Paleomagnetic poles of Pampia terrane considered in the paleogeographic analysis. All the poles were rotated to African coordinates according to Euler's pole: 47.5° N, 33.3° W, R: 56.2° (Torsvik et al., 2012). Pole references: 1- This paper, 2- Franceschinis et al., 2016, 3- Spagnuolo et al., 2012, 4- Rodríguez Piceda et al., 2018. Age references: a- Aparicio González et al., 2011, b- Escayola et al., 2011, c- This paper, d- Franceschinis et al., 2016, e- Augustsson et al., 2011, f- Escayola (com. personal), g- Spagnuolo et al., 2012; h- Rodríguez Piceda et al., 2018. In italics, data that may not have averaged out full paleosecular variation. In bold, data presented in this paper.

Geologic unit	Pole	Present-day coordinates		Pole position (African coordinates)		A95 (°)	Q	Age (method)	Ref.
		Lat (°)	Long (°)	Lat (°)	Long (°)				
Mojotoro Granite	***MO1*** (1)	***9.1***	***345.6***	***30.8***	***17.1***	***16.4***	***1010000***	***533 ± 2 (U-Pb LA-ICP-MS, in zircon)***	***a***
Puncoviscana Formation	***TP1*** (1)	***34.5***	***52.3***	***68.0***	***118.4***	***5.0***	***1010000***	***536 ± 5 (U-Pb SHRIMP, in zircon)***	***b***
Puncoviscana Formation	***TP2*** (1)	***38.3***	***33.6***	***75.4***	***73.2***	***5.7***	***1010000***	***< 536 (stratigraphy)***	***c***
Puncoviscana Formation	*P1* (2)	*38.9*	*355.5*	*59.5*	*359.4*	*4.5*	*0010000*	*< 530 (stratigraphy)*	*d*
Puncoviscana Formation	*P2* (2)	*21.2*	*275.2*	*0.4*	*312.0*	*7.2*	*1010000*	*537 ± 1 (U-Pb SHRIMP, in zircon)*	*b*
Campanario Formation (Terma de Reyes)	***R*** (1)	***23.4***	***349.5***	***44.7***	***10.6***	***2.5***	***0110001***	***514–485***	***c***
Campanario Formation (Tilcara)	***T*** (1)	***33.3***	***341.6***	***47.5***	***353.6***	***2.8***	***1111011***	***< 514 ± 5 (U-Pb LA-ICP-MS, detrital zircons)***	***e***
Campanario Formation (Santa Victoria Oeste)	*M* (2)	*18.3*	*358.8*	*45.4*	*24.9*	*3.9*	*1110001*	*< 519 ± 4.8 (U-Pb, detrital zircons)*	*f*
Campanario Formation (Mean pole)	**C2** (1)	**23.6**	**346.5**	**43.2**	**7.5**	**7.0**	**1111111**	**514–485 (detrital zircons, stratigraphy)**	**c**
Campanario Formation (Matancillas)	C (3)	−1.9	1.0	28.2	38.6	4.0	0111011	514–485	g
La Pedrera Formation	LP (4)	38.3	340.4	50.1	347.0	8.8	0111011	485–477 (fossil fauna)	h
Santa Rosita Formation	SR (3)	8.6	355.3	35.2	27.2	10.1	0111001	485–477 (fossil fauna)	g

formation (sites TP1 and TP2) and the Mojotoro intrusion (MO) did not average the paleosecular variation and should be considered as “spot readings” of the Early Cambrian paleomagnetic field.

The available Cambrian and Ordovician paleomagnetic poles and virtual geomagnetic poles (VGPs) for the Pampia terrane rotated to North African coordinates (Torsvik et al., 2012) in a Gondwana reconstruction are depicted in Fig. 14 and presented in Table 4. They include those presented in this paper and those previously published by Spagnuolo et al. (2008, 2012), and Rodríguez Piceda et al. (2018). These previous works computed paleomagnetic pole for the Campanario Formation (late Middle to early Late Cambrian) and two Early Ordovician Santa Rosita and La Pedrera Formations (Santa Victoria Group), respectively. Based on the distribution of poles from the Campanario and Santa Rosita Formations, Spagnuolo et al. (2012) and Franceschinis et al. (2016) cautiously favored a tectonic model in which Pampia was accreted to the SW Gondwana margin by a strike-slip displacement along the margin of the Río de la Plata craton from a position close to the Kalahari craton, during the Middle to Late Cambrian. This model followed a previous one by Rapela et al. (2007) but prolonged the displacement up to the end of the Cambrian. From our study and previous results from Franceschinis et al. (2016), a grand mean Campanario pole based on data from four localities has been computed. With the new information, a re-evaluation of the proposed models may be pertinent. The most direct way to determine paleomagnetically the relative kinematic evolution of Pampia with respect to other Gondwana blocks in Cambrian times is through the comparison of coeval pole positions from these blocks in different paleogeographic reconstructions. Unfortunately, there is no reliable and well dated paleomagnetic information from the Río de la Plata craton for the Cambrian (Rapalini, 2018), just as there is no data on Neoproterozoic units of the Pampia terrane. Therefore, a strict comparison between both adjacent blocks is still impossible. The analysis must resort, then, to pole positions obtained from other Gondwana cratons, considering that they were already part of the supercontinent and did not suffer significant relative displacements after the Cambrian (while they were part of Gondwana). In this case, the potential information corresponding to the Congo-São Francisco craton appears as relevant. According to paleomagnetic (Rapalini, 2018 and references therein) and geological information (Oriolo et al., 2017 and references therein), this craton and Río de la Plata were already attached by *ca.* 575 Ma. Therefore, the same apparent polar wander path (APWP) must apply for both cratons since the Middle Ediacaran (Fig. 14). The paleomagnetic pole of the Cambrian Piquete Formation (*ca.*500 Ma, PF, D'Agrella Filho et al., 1986) is the only one for such craton that may be comparable to that of the Campanario Fm. However, this is not a particularly robust pole. It was computed from metamorphic rocks and its age is not precisely determined (D'Agrella Filho et al., 2000).

Considering that a comparison with Congo- São Francisco poles is hampered by the small number of pole positions available for this block, in Fig. 14, we plotted the Cambrian to Ordovician paleomagnetic poles for Pampia in comparison with mean reference poles for the whole of Gondwana, following similar analysis by Spagnuolo et al. (2012), Franceschinis et al. (2016) and Rodríguez Piceda et al. (2018). Unfortunately, there is no consensus in the calculation of the Early Paleozoic reference poles for this supercontinent (see, for example, Rodríguez Piceda et al., 2018). In this case, the mean poles computed by Torsvik et al. (2012) from a systematic compilation and analysis of Cambro-Ordovician poles from all Gondwana blocks is used. There is a strong tendency of the Pampia terrane late Cambrian-Early Ordovician poles to fall in the extreme north of the African continent, only partially overlapping or removed from the mean paleomagnetic poles of Gondwana for 530, 520, 510, 500, 490 and 480 Ma (Fig. 15).

Values of rotation and inclination anomaly (flattening), and their uncertainties, were calculated for each paleomagnetic pole of Pampia with respect to the reference poles of closest age (Demarest, 1983; Beck, 1989). These results are represented in Fig. 16. The distribution of these values with respect to the age of the rocks from which they were obtained can provide relevant independent information on Pampia's

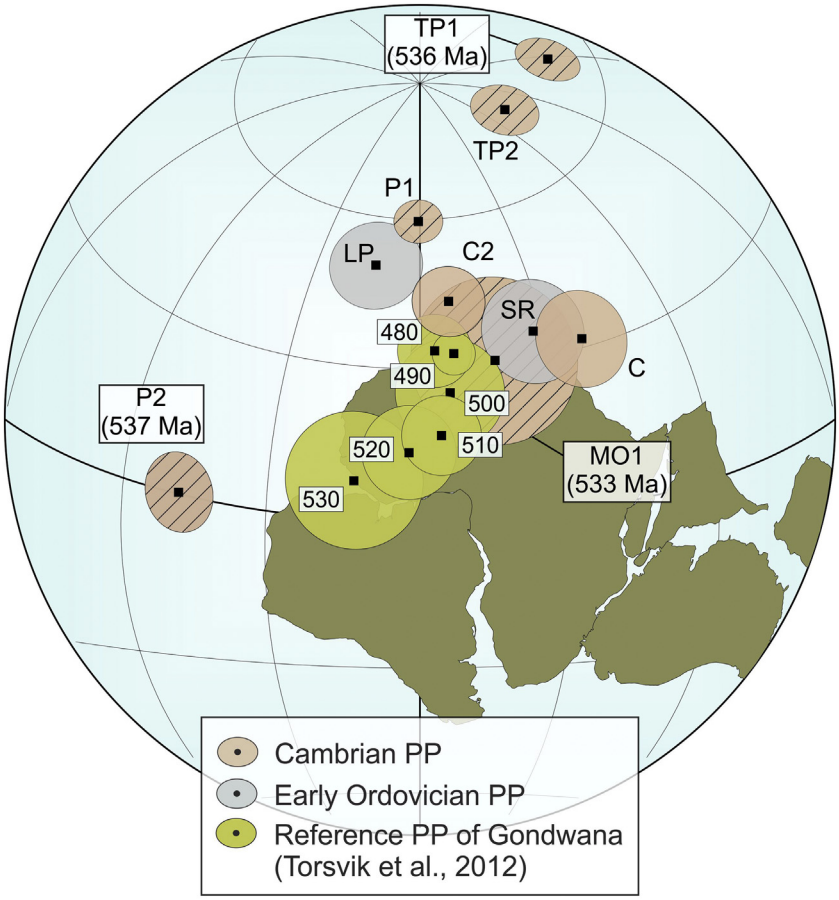


Fig. 15. Paleomagnetic poles of the Pampia terrane compared with the mean paleomagnetic poles of Gondwana for the 530, 520, 510, 500, 490 and 480 Ma (Torsvik et al., 2012) The Pampia poles and VGPs were rotated to African coordinates according to Torsvik et al. (2012). VGPs are indicated with a striped plot.

lateral displacement kinematic model (Spagnuolo et al., 2012; Franceschinis et al., 2016). The Pampia drift model (Fig. 17) from the southern margin of the Kalahari craton to the western margin of the Río de la Plata block involves a major dextral strike-slip movement of about 2000 km (Spagnuolo et al., 2012). During such displacement, Pampia must have displaced latitudinaly towards the north (present day coordinates, approximately to the south according to the paleogeographic orientation of Gondwana) as well as rotated clockwise.

Franceschinis et al. (2016) suggested that the different locations of the paleomagnetic poles of the Campanario Formation computed from outcrops at Matancillas-Iruya and Santa Victoria Oeste might be reflecting different stages of Pampia displacement along the Rio de la Plata craton margin (Fig. 17). With the new poles obtained in our study, and that recently published by Rodríguez Piceda et al. (2018), a more reliable paleomagnetic test of such model can be attempted. According to the invoked kinematics of Pampia during Cambrian times there should be some correlation between a larger inclination anomaly (flattening) and a higher degree of clockwise rotation, decreasing both gradually until Pampia reached a position similar to the current one with respect to the Río de la Plata craton by latest Cambrian – earliest Ordovician (Spagnuolo et al., 2012). Therefore, this model imposes a chronological progression (larger rotation and inclination anomaly in older units) but also some kind of proportional relationship between both parameters. This should produce a direct relationship in the diagram of Fig. 16 between F and R for the Campanario Fm. (late middle to late Cambrian) and the Santa Rosita and La Pedrera Formations (Early Ordovician). Not such pattern is observed. Furthermore, both the Cambrian and Ordovician poles obtained at the Matancillas-Iruya areas show a significant clockwise rotation, with much smaller to non-significant rotation values in the case of the Campanario mean (all other localities) and the La Pedrera pole. Furthermore, the relative positions of these Late Cambrian and Early Ordovician poles of Pampia is virtually identical to that found in the Matancillas-Iruya area. This is very clearly illustrated in Fig. 16c in which the Campanario and Santa Rosita poles obtained in the Matancillas-Iruya area have been rotated 30° counterclockwise around a vertical axis in the sampling locality. Consistency of both, the late Cambrian and early Ordovician poles from the different localities strongly suggest that the "anomalous" position of Cambrian and Ordovician poles found by Spagnuolo et al. (2008, 2012) is due to local tectonic rotations. Spagnuolo et al. (2008) suggested that those localities might be located in a dextral transfer shear zone along the Andean orogenic front and that might underwent around 30° of *in situ* clockwise rotation in Late Cenozoic times. Fig. 1c shows that both localities, Matancillas and Iruya (Parada del Condor), are at approximately the same latitude (near 23°S). At this latitude, the Andean orogenic front experience a major change from a "thick-skinned" deformation style to the south (Santa Barbara System, Kley and Monaldi, 2002) with Andean tectonic shortening in the order of 25–30 km to the "thin-skinned" deformation style of the Sub-Andean Ranges, in which shortening or around 100 km have been calculated (Kley and Monaldi, 1998, 2002). The transition between both areas of the orogenic front does not correspond to any specific transversal fault system, and has been interpreted as a zone of diffuse dextral wrench deformation. Andean deformation along this transfer zone may have promoted *in situ* clockwise rotations as observed in Matancillas and Iruya localities. Unfortunately, paleomagnetic data on younger rocks in the area are scarce. Maffione et al. (2009) reported paleomagnetic data on Cretaceous to Neogene sedimentary rocks of the Eastern Cordillera and Puna and found clockwise rotations of around 45° to 15°. Although some of these data show large angular uncertainties and belong to outcrops a

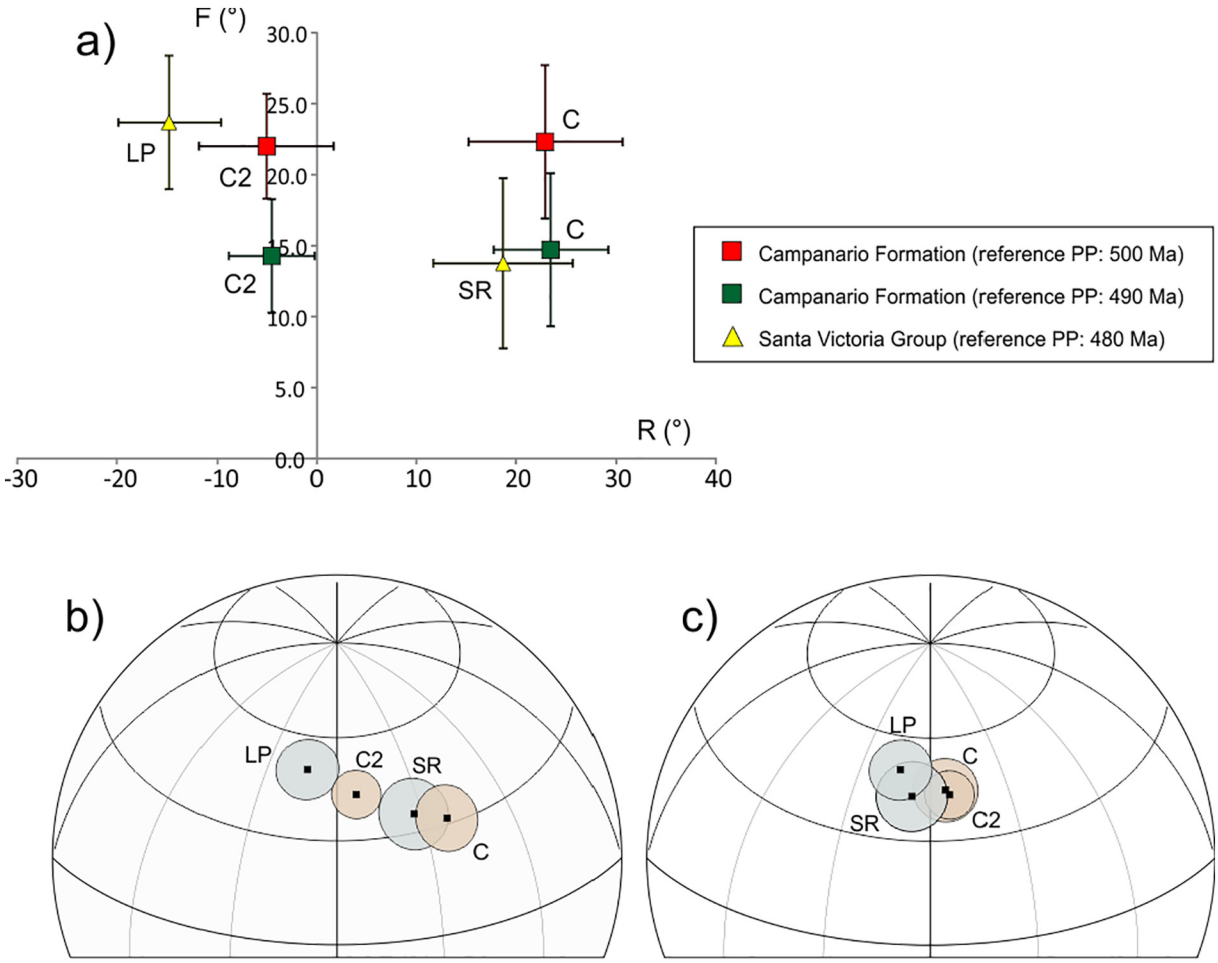


Fig. 16. a) F (anomaly in inclination) *vs.* R (rotation) calculated for the paleomagnetic poles of Campanario Formation (square), and the Santa Victoria Group (Santa Rosita and La Pedrera Formations, yellow triangles) using the reference paleomagnetic poles of Gondwana for 500, 490, 480 Ma (Torsvik et al., 2012). Note the presence of some counterclockwise rotations; b) poles of the Campanario formation (C and C2) and Santa Victoria Group (LP and SR), based on site mean directions, rotated to African coordinates according to Euler pole 47.5° N, 33.3° W, 56.2° ccw (Torsvik et al., 2012); c) idem b) with poles of the Campanario formation (C) and the Santa Victoria Group (SR) determined by Spagnuolo et al. (2012) rotated 30° ccw around an Euler's pole 24.05° S, 354.7° W. Note agreement of late Cambrian and early Ordovician poles after rotation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

few tens of kilometers to the west of the Iruya locality, they are concentrated along a stripe at the same latitude (around 22°45′S) as the two localities studied by Spagnuolo et al. (2008, 2012). Although Maffione et al. (2009) extrapolated their data to a much larger region (Eastern Cordillera between 22 and 24°S), almost all paleomagnetic data come from such strip. This clearly supports the simplest explanation that only the Iruya-Matancillas area has been affected by large Andean clockwise rotations and that the mean paleomagnetic pole computed for the Campanario Formation by averaging data from four different localities is a more reliable Late Cambrian paleomagnetic pole for Pampia. The Early Ordovician pole position for this terrane is probably best represented by the La Pedrera pole (Rodríguez Piceda et al., 2018), obtained far away from the rotated area. Following this, our new data clearly supports the hypothesis of a local Andean rotation of the Early Paleozoic outcrops in the Iruya-Matancillas area (Spagnuolo et al., 2008) against the alternative interpretation of a recording of strike-slip displacement of Pampia in late Cambrian times. These observations are, therefore, not consistent with the tectonic models proposed by Spagnuolo et al. (2012) and Franceschinis et al. (2016). Rapela et al. (2007) suggested that displacement of Pampia occurred somewhat earlier (Early Cambrian). Our paleomagnetic data for such ages is still scarce and of low reliability for a meaningful test of that proposal. The palelaotitude anomaly observed by our data from Pampia when compared with the Gondwana reference poles (Torsvik et al., 2012) could be interpreted as evidence of a younger displacement along the Gondwana margin, but geological evidence strongly opposes such scenario (see Rapela et al., 2007; Ramos et al., 2010, and references therein). Large underestimation of inclination shallowing in the clastic sedimentary rocks of the Campanario, Santa Rosita and La Pedrera Formations seem unlikely at a first glance, as experimental tests of remanence anisotropy were performed in most of these rocks. However, it is worth keeping in mind that anisotropy tests performed on these rocks were obtained with fields up to 1 T and may not be fully representative for hematite-bearing sandstones. On the other hand, whether a systematic bias affect the Late Cambrian – Early Ordovician Gondwana mean poles also needs some more scrutiny but will not be solved without new high-quality and well-dated paleomagnetic poles from stable areas of South America and other Gondwana forming continents, as recently analyzed by Rodríguez Piceda et al. (2018), who found significant changes when comparing the Early Ordovician Pampia pole with different reference paths for Gondwana.

10. Conclusions

A regional paleomagnetic sampling was carried out in Cambrian units of the Pampia terrane. These comprised sites at three new localities in the Campanario Formation (Middle to Late Cambrian), in the Early Cambrian Mojotoro intrusive and in the Puncoviscana Formation.

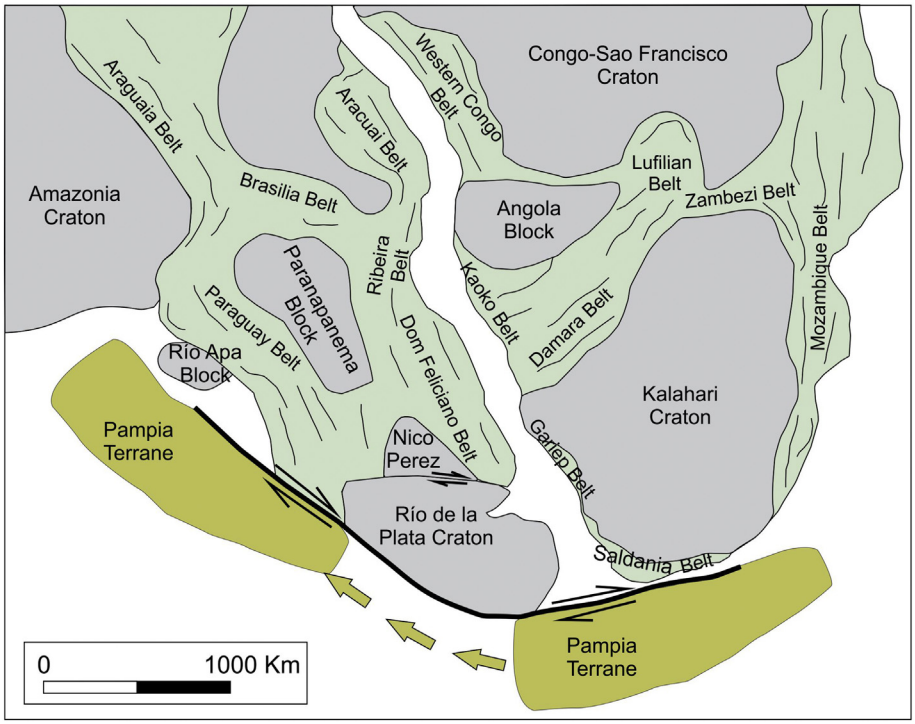


Fig. 17. Model of strike-slip dextral displacement of Pampia proposed by Spagnuolo et al. (2012) for assembly of this terrane to Gondwana in Late Cambrian times (see discussion and references in the text). Modified from Rapela et al. (2011).

The new paleomagnetic poles obtained in the Campanario Formation, and previous ones in the same formation and the Early Ordivician Santa Rosita and La Pedrera Formations show inclination anomalies and rotations inconsistent with the model of a dextral displacement of the Pampia terrane from the Kalahari craton in Middle to Late Cambrian times. An overall mean paleomagnetic pole for the Campanario Formation, computed on a mean-site basis from four different localities, is the most likely best representation of the Pampia pole position in the Late Cambrian. This leaves out data coming from the Iruya-Matancillas area that has been likely affected by near 30° of *in situ* clockwise rotation due to Andean tectonics. This rotation also affected the Early Ordovician Santa Rosita Formation. The virtual geomagnetic poles obtained from the Early Cambrian Puncoviscana Formation are highly anomalous with respect to the accepted Gondwana apparent polar wander path as well as with previous VGPs obtained from this formation in another locality. The VGP from the early Cambrian Mojotoro Granite, instead, falls close to the Campanario poles.

CRediT authorship contribution statement

Pablo R. Franceschinis: Investigation, Conceptualization, Methodology, Formal analysis, Writing - original draft, Visualization. **Augusto E. Rapalini:** Investigation, Conceptualization, Methodology, Writing - review & editing, Visualization, Project administration, Funding acquisition. **Mónica P. Escayola:** Investigation, Conceptualization, Writing - review & editing, Funding acquisition. **Constanza Rodríguez Piceda:** Investigation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by the Universidad de Buenos Aires (Grant 20020170100290BA) and CONICET (Subsidio Unidad Ejecutora-IGEBA 2016). Super IAPD and GMAP software packages by Torsvik (NGU) and Remasoft program (AGICO SA) were used to analyse the data.

Fernando Almaraz and Matias Naselli kindly collaborated during preparation of samples and running of thermomagnetic curves. Discussions of some rock magnetic results are also acknowledged.

The manuscript was greatly improved by thorough but constructive reviews by two anonymous reviewers and suggestions by Associate Editor Zheng Xiang Li.

References

Adams, C.J., Miller, H., Aceñolaza, F.G., Toselli, A.J., Griffin, W.L., 2011. The Pacific Gondwana margin in the late Neoproterozoic-early Paleozoic: detrital zircon U-Pb ages from metasediments in northwest Argentina reveal their maximum age, provenance and tectonic setting. Gondwana Res. 19, 71–83. https://doi.org/10.1016/j.gr.2010.05.002.

Aparicio González, P.A., Pimentel, M.M., Hauser, N., 2011. Datación U-PB por LA-ICP-MS de diques graníticos del ciclo pampeano, sierra de Mojotoro, Cordillera Oriental. Revista de la Asociacion Geologica Argentina 68, 33–38.

Augustsson, C., Rusing, T., Adams, C.J., Chmiel, H., Kocabayoglu, M., Buld, M., Zimmermann, U., Berndt, J., Kooijman, E., 2011. Detrital quartz and zircon combined: the production of mature sand with short transportation paths along the Cambrian West Gondwana margin, Northwestern Argentina. J. Sediment. Res. 81, 284–298. https://doi.org/10.2110/jsr.2011.23.

Beck, M.E., 1989. Paleomagnetism of continental North America; implications for displacement of crustal blocks within the Western Cordillera, Baja California to British Columbia. Geological Society of America Memoirs 172, 471–492.

Buatois, L.A., Mángano, M.G., 2003. La icnofauna de la Formación Puncoviscana en el noroeste argentino: La colonización de fondos oceánicos y reconstrucción de paleoambientes y paleoecosistemas de la transición precámbrica-cámbrica. Ameghiniana 40, 103–117.

Cordani, U.G., Pimentel, M.M., De Araújo, C.E.G., Basei, M.A.S., Fuck, R.A., Girardi, V.A.V., 2013. Was there an Ediacaran Clymene ocean in central South America? Am. J. Sci. 313, 517–539.

D'Agrella Filho, M.S., Pacca, I.G., Sato, K., 1986. Paleomagnetism of metamorphic rocks from the Piquete region-Ribeira Valley, Southeastern Brazil. Revista Brasileira de Geofísica 4, 79–84.

D'Agrella Filho, M.S., Babinski, M., Trindade, R.I., Van Schmus, W.R., Ernesto, M., 2000.

Simultaneous remagnetization and U–Pb isotope resetting in Neoproterozoic carbonates of the Sao Francisco craton, Brazil. Precambrian Res. 99, 179–196.
Demarest, H., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res. 88, 4321–4328.
Escayola, M.P., Pimentel, M.M., Armstrong, R., 2007. Neoproterozoic backarc basin: Sensitive high-resolution ion microprobe U-Pb and Sm-Nd isotopic evidence from the Eastern Pampean Ranges, Argentina. Geology 35, 495–498. https://doi.org/10.1130/G23549A.1.
Escayola, M.P., van Staal, C.R., Davis, W.J., 2011. The age and tectonic setting of the Puncoviscana Formation in northwestern Argentina: an accretionary complex related to Early Cambrian closure of the Puncoviscana Ocean and accretion of the Arequipa-Antofalla block. J. S. Am. Earth Sci. 32, 438–459. https://doi.org/10.1016/j.jsames.2011.04.013.
Franceschinis, P.R., Rapalini, A.E., Escayola, M.P., Luppo, T., 2016. Paleomagnetic studies on the late Ediacaran - Early Cambrian Puncoviscana and the late Cambrian Campanario formations, NW Argentina: New paleogeographic constraints for the Pampia terrane. J. S. Am. Earth Sci. 70, 145–161. https://doi.org/10.1016/j.jsames.2016.04.007.
Gosen, W. Von, Prozzi, C., 2010. Pampean deformation in the Sierra Norte de Córdoba, Argentina: implications for the collisional history at the western pre-Andean Gondwana margin. Tectonics 29, 1–33. https://doi.org/10.1029/2009TC002580.
Hodych, J.P., Buchan, K.L., 1994. Early Silurian palaeolatitude of the Springdale Group redbeds of central Newfoundland: a palaeomagnetic determination with a remanence anisotropy test for inclination error. Geophys. J. Int. 117, 640–652. https://doi.org/10.1111/j.1365-246X.1994.tb04034.x.
Hrouda, F., Verner, K., Kubínová, Š., Buriánek, D., Wali, S., Chlupáčová, M., Holub, F.V., 2016. Magnetic fabric and emplacement of dykes of lamprophyres and related rocks of the Central Bohemian Dyke Swarm (Central European Variscides). J. Geosci. 61, 335–354. https://doi.org/10.3190/jgeosci.222.
Jelinek, V., 1981. Characterization of the magnetic fabrics of rocks. Tectonophysics 79, T63–T67.
JeŽek, P., Willner, A.P., Aceñolaza, F.G., Miller, H., 1985. The Puncoviscana trough - a large basin of Late Precambrian to Early Cambrian age on the pacific edge of the Brazilian shield. Geol. Rundsch. 74, 573–584. https://doi.org/10.1007/BF01821213.
Keidel, J., 1943. El Ordovícico inferior de los Andes del norte argentino y sus depósitos marinos-glaciares. 36. Academia Nacional de Ciencias, Córdoba, Argentina, pp. 140–229.
King, R.F., 1955. The Remanent Magnetism of Artificially Deposited Sediments. Geophysical Supplements to the Monthly Notices of the Royal Astronomical Society 7, 115–134.
Kirschvink, J., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. Int. 62, 699–718.
Kley, J., Monaldi, C.R., 1998. Tectonics shortening and crustal thickness in the Central Andes: how good is the correlation? Geology 26, 723–726.
Kley, J., Monaldi, C.R., 2002. Tectonic inversion in the Santa Barbara System of the central Andean foreland thrust belt, northwestern Argentina. Tectonics 21, 11.
Kodama, K.P., 2012. Paleomagnetism of Sedimentary Rocks: Processes and Interpretation. Wiley-Blackwell (157 pp.).
Koymans, M.R., Langereis, C.G., Pastor-galán, D., Van Hinsbergen, D.J.J., 2016. Paleomagnetism.org: an online multi-platform open source environment for paleomagnetic data analysis. Comput. Geosci. 93, 127–137. https://doi.org/10.1016/j.cageo.2016.05.007.
Kraemer, P.E., Escayola, M.P., Martino, R.D., 1995. Hipótesis sobre la evolución tectónica neoproterozoica de las Sierras Pampeanas de Córdoba (30° 40′–32° 40′), Argentina. Rev. Asoc. Geol. Argent. 50, 1–4.
Maffione, M., Speranza, F., Faccenna, C., 2009. Bending of the Bolivian orocline and growth of the central Andean plateau: paleomagnetic and structural constraints from the Eastern Cordillera (22–24 S, NW Argentina). Tectonics 28, 4.
Moya, M.C., 1998. El Paleozoico Inferior de la sierra de Mojotoro, Salta-Jujuy. Rev. Asoc. Geol. Argent. 53, 219–238.
Oriolo, S., Oyhantçabal, P., Wemmer, K., Siegesmund, S., 2017. Contemporaneous assembly of Western Gondwana and final Rodinia break-up: implications for the supercontinent cycle. Geosci. Front. 8, 1431–1445. https://doi.org/10.1016/j.gsf.2017.01.009.
Pares, J.M., 2015. Sixty years of anisotropy of magnetic susceptibility in deformed sedimentary rocks. Frontiers in Earth Sciences 3, 1–13. https://doi.org/10.3389/feart.2015.00004.
Ramos, V.A., Vujovich, G., Martino, R., Otamendi, J., 2010. Pampia: a large cratonic block missing in the Rodinia supercontinent. J. Geodyn. 50, 243–255. https://doi.org/10.1016/j.jog.2010.01.019.
Rapalini, A.E., 2018. The assembly of Western Gondwana: reconstruction based on paleomagnetic data. In: Siegesmund, S., Basei, M.A.S., Oyhantçabal, P., Oriolo, S. (Eds.), Geology of Southwest Gondwana. Regional Geology Reviews. Springer International Publishing, Cham, pp. 3–18. https://doi.org/10.1007/978-3-319-68920-3.
Rapalini, A.E., Tohver, E., Sánchez, L., Lossada, A.C., Barcelona, H., Pérez, C., 2015. The late Neoproterozoic Sierra de las Ánimas Magmatic Complex and Playa Hermosa Formation, southern Uruguay, revisited: paleogeographic implications of new paleomagnetic and precise geochronologic data. Precambrian Res. 259, 143–155. https://doi.org/10.1016/j.precamres.2014.11.021.
Rapela, C.W., Pankhurst, R.J., Saavedra, J., Galindo, C., Casquet, C., Baldo, E.G.A., 1998. Early evolution of the proto-andean margin of South America. Geology 26, 707–710. https://doi.org/10.1130/0091-7613(1998)026<0707:EEOTPA>2.3.CO.
Rapela, C.W., Pankhurst, R.J., Casquet, C., Fanning, C.M., Baldo, E.G.A., González-Casado, J.M., Galindo, C., Dahlquist, J., 2007. The Río de la Plata craton and the assembly of SW Gondwana. Earth Sci. Rev. 83, 49–82. https://doi.org/10.1016/j.earscirev.2007.03.004.
Rapela, C.W., Fanning, C.M., Casquet, C., Pankhurst, R.J., Spalletti, L., Poiré, D., Baldo, E.G., 2011. The Rio de la Plata craton and the adjoining Pan-African/brasiliano terranes: their origins and incorporation into south-west Gondwana. Gondwana Res. 20, 673–690. https://doi.org/10.1016/j.gr.2011.05.001.
Rapela, C.W., Verdecchia, S.O., Casquet, C., Pankhurst, R.J., Baldo, E.G.A., Galindo, C., Murra, J.A., Dahlquist, J.A., Fanning, C.M., 2015. Identifying Laurentian and SW Gondwana sources in the Neoproterozoic to Early Paleozoic metasedimentary rocks of the Sierras Pampeanas: paleogeographic and tectonic implications. Gondwana Res. 32, 193–212. https://doi.org/10.1016/j.gr.2015.02.010.
Robert, B., Besse, J., Blein, O., Greff-Lefftz, M., Baudin, T., Lopes, F., Meslouh, S., Belbadaoui, M., 2017. Constraints on the Ediacaran inertial interchange true polar wander hypothesis: a new paleomagnetic study in Morocco (West African Craton). Precambrian Res. 295, 90–116.
Rodríguez Piceda, C., Franceschinis, P.R., Escayola, M.P., Rapalini, A.E., 2018. Paleomagnetismo del Grupo Santa Victoria en la sierra de Mojotoro, Salta: aportes a la reconstrucción paleogeográfica de Pampia en el Paleozoico Temprano. Revista de la Asociacion Geologica Argentina. 4. pp. 518–532. https://doi.org/10.1016/j.psep.2018.03.001.
Ruiz Huidobro, O.J., 1968. Descripción geológica de la hoja 7e Salta (Provincias de Salta y Jujuy). Carta Geológico-Económica de la República Argentina, escala 1:200.000. Instituto Nacional de Geología y Minería, Buenos Aires.
Schwartz, J.J., Gromet, L.P., 2004. Provenance of a late Proterozoic-early Cambrian basin, Sierras de Córdoba, Argentina. Precambrian Res. 129, 1–21. https://doi.org/10.1016/j.precamres.2003.08.011.
Spagnuolo, C.M., Rapalini, A.E., Astini, R.A., 2008. Paleogeographic and tectonic implications of the first paleomagnetic results from the Middle-Late Cambrian Mesón Group: NW Argentina. J. S. Am. Earth Sci. 25, 86–99. https://doi.org/10.1016/j.jsames.2007.08.004.
Spagnuolo, C.M., Rapalini, A.E., Astini, R.A., 2012. Assembly of Pampia to the SW Gondwana margin: a case of strike-slip docking? Gondwana Res. 21, 406–421. https://doi.org/10.1016/j.gr.2011.02.004.
Tan, X., Kodama, K.P., 2002. Magnetic anisotropy and paleomagnetic inclination shallowing in red beds: evidence from the Mississippian Mauch Chunk Formation, Pennsylvania. Journal of Geophysical Research: Solid Earth 107, EPM 9–1-EPM 9-17. https://doi.org/10.1029/2001JB001636.
Tauxe, L., Watson, G.S., 1994. The fold test: an eigen analysis approach. Earth Planet. Sci. Lett. 122, 331–341.
Tohver, E., Trindade, R.I., 2014. Comment on “Was there an Ediacaran Clymene Ocean in central South America?” by UG Cordani and others. Am. J. Sci. 3143, 805–813.
Torsvik, T.H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P.V., Douwe, van Hinsbergen, D.J.J., Domeier, M., Gaina, C., Tohver, E., Meert, J.G., McCausland, P.J.A., Cocks, L.R.M., 2012. Phanerozoic polar wander, palaeogeography and dynamics. Earth Sci. Rev. 114, 325–368.
Toselli, A.J., Alonso, R., 2005. Pórfiro granítico Mojotoro (Salta): ¿Una cúpula intrusiva o un dique en el ciclo pampeano? In: Revista de la Asociación Geológica Argentina. 2. pp. 428–430.
Trindade, R.I.F., D’Agrella Filho, M.S., Epof, I., Brito-Neves, B.B., 2006. Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana. Earth Planet. Sci. Lett. 244, 361–377. https://doi.org/10.1016/j.epsl.2005.12.039.
Turner, J.C., 1964. Descripción Geológica de la Hoja 2c Santa Victoria (Provincias de Salta y Jujuy). Carta geológica-económica de la República Argentina, escala 1:200.000. Servicio Geológico Minero Argentino, Buenos Aires.
Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.
Worm, H.-U., Jackson, M., 1999. The superparamagnetism of Yucca Mountain Tuff. Journal of Geophysical Research: Solid Earth 104, 25415–25425. https://doi.org/10.1029/1999JB900285.